DATA-DRIVEN WILDFIRE SPREAD MODELING USING EARTH OBSERVATION DATA WITH INTERACTIVE 3D VISUALIZATION IN UNITY

PARIS-LODRON UNIVERSITY SALZBURG (PLUS)

Faculty of Digital and Analytical Sciences

Department of Geoinformatics

And

PALACKÝ UNIVERSITY OLOMOUC (UPOL)

Faculty of Science

Department of Geoinformatics

PLUS supervisor: PhD Karima Hadj-Rabah

UPOL supervisor: PhD Jan Brus

submitted by

Marcelo Rigoto

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Hereby, I declare that this piece of work is entirely my own, the references cited have been acknowledged, and the thesis has not been previously submitted to the fulfilment of the higher degree. During the preparation of this thesis, generative artificial intelligence tools were used. Their use is described in Section 3.



I would like to thank my supervisors for their help. For the staff in Salzburg and Olomouc for the teachings. I would like to dedicate this thesis to my parents, Claudicea Rigoto and Marcos Rigoto, for always believing in me and investing in my education, and allowing me to get where I am. I would also like to thank all my friends that I made during the Master's, especially Bruna and Bia. Lastly, I would like to thank Maria for the partnership and help through the hard times of the thesis.

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ASSIGNMENT OF DIPLOMA THESIS

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Personal number:	R241012
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Theses guidelines

The aim of this thesis is to design and implement a software tool in the Unity engine for automated 3D visualization of wildfire spread and risk areas using real-world Earth Observation data. The core of the solution is the development of a data pipeline capable of extracting, processing, and importing geographic datasets, specifically digital elevation models, satellite imagery, and fire-related products with a focus on ESA CCI Fire, Copernicus EFFIS, or FIRMS databases. The student will focus on the algorithmic conversion of GIS data into 3D space and their integration into scene and rendering workflows, where the input dataset automatically defines the terrain topology, surface textures, and wildfire parameters. The resulting data pipeline will allow for the setup of an interactive tool capable of dynamically generating a scene of a selected location based on external data inputs and visualizing wildfire conditions corresponding to near real-time or historical situations. The student will create a website about the thesis following the rules available on the department's website and a poster about the diploma thesis in A2 format. The student will submit the entire text (text, attachments, poster, outputs, input and output data) in digital form on a storage medium to the department's secretary

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Supervisors of diploma thesis: RNDr. Jan Brus, Ph.D.
Department of Geoinformatics

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L.S.

prof. RNDr. Martin Kubala, Ph.D.
Dean

prof. RNDr. Vilém Pechanec, Ph.D.
Head of Department

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Content

Content 11
1. Introduction 15
1.1 The Meaning of Modeling in this Thesis 17
1.2 Research Problem and Motivation 18
1.3 Objectives 18
1.4 Contributions 19
2. Background and Literature Review 20
2.1 Wildfire Ecology and the Growth of Anthropogenic Threat 20
2.2 Satellite-Based Fire Monitoring: NASA FIRMS 20
2.3 Existing Wildfire Visualization and Simulation Tools 21
2.4 Platform Selection: Alternatives Considered 22
2.5 Game Engines for Geospatial Visualization 27
2.6 Narrative Cartography and Immersive Spatio-temporal Storytelling 28
3. Methodology 30
3.1 Workflow Overview 30
3.2 Study Area: Jasper National Park, Alberta, Canada (2024) 32
3.3 Input Data 32
3.3.1 Fire Detection Data 33
3.3.2 Meteorological Data 33
3.3.3 Terrain and Imagery 33
3.4 Cesium Coordinate System 34
3.5 Data Acquisition Pipeline 35
3.6 Temporal Animation 36

3.7 Weather Integration and Visualization Overlays 37
3.8 User Interface and Interaction Design 38
3.9 Unity Development 38
3.9.1 Desktop deployment 39
3.9.2 WebGL Deployment and Browser Execution 41
4.1 Aims of the User Testing 43
4.2 Participants 44
4.3 Materials: Three Tools and the Three Scenarios 46
4.3.1 Selection of the Comparison Tools 46
4.3.2 Scenario Selection and Comparability 47
4.4 The Questionnaire Instrument 49
4.5 Procedure and Online Delivery 50
4.6 Data Analysis Plan 51
4.7 Design Limitations and Ethical Considerations 53
5. Results 54
5.1 Comprehension 54
5.2 Usability and Ratings 55
5.3 Influence of Participant Background 59
5.4 Open-Ended Responses 62
5.5 Comparison of the Desktop and Browser Builds 64
6. Discussion 67
6.1 Objectives Revisited 67
6.2 Principal Findings 68
6.3 Related Work and Limitations 70
7. Conclusion 71
References 75

Figures.....75
Literature.....75
List of Attachmets.....79
Appendix.....80
Appendix A: Supporting Study Data.....80

Abstract

Wildfire seasons have grown longer, more frequent and more intense in recent decades, demanding better tools for public awareness and decision support. Commonly used GIS dashboards and scientific simulators provide rich data but commonly lack the narrative clarity demanded to engage non-technical audiences. This thesis presents a purely visual, time-lapse wildfire visualization pipeline built in Unity3D. The modular framework ingests Cesium elevation models and NASA FIRMS active fire detections, converting them into georeferenced terrain, draped imagery and particle-based fire emitters. Detections are clustered within a user-configured spatial radius, marker size is scaled by fire radiative power to convey relative intensity, and all layers are sequenced on a day-by-day timeline the viewer can scrub interactively. Key parameters are exposed as editable settings, so the workflow can be adapted to other events without code changes. The application is delivered as a desktop reference build and as a browser deployment of the same pipeline requiring no installation. The browser deployment was evaluated against kepler.gl and an ArcGIS Online Experience Builder application in a counterbalanced study with 68 participants. It produced the highest mean comprehension score of the three tools and was rated as favorably as the commercial platform on the visualization qualities assessed, while being found significantly less usable, a gap concentrated within participants without prior three-dimensional or GIS experience. The strongest result concerns terrain: participants using the tool without a rendered elevation surface described a mountainous landscape correctly in 9.5 percent of cases, against near-complete accuracy for both terrain-based tools, and every incorrect answer was confident rather than uncertain. The pipeline requires no paid license and no proprietary GIS platform.

1. Introduction

The increasing severity, frequency and geographic extent of wildfires have made communicating their impacts to the public and decision-makers one of the major challenges in contemporary environmental science. Recent decades have seen an exponential increase in wildfire season length, frequency and total burned area (Figure 1), driven largely by warmer temperatures, longer dry seasons and more severe droughts associated with climate change (United States Environmental Protection Agency, 2022), a trend confirmed independently by a century-long reconstruction of global burned area (Guo et al., 2025). In addition to these overall trends, the most extreme wildfire events, defined by exceptionally high Fire Radiative Power (FRP), have become increasingly common, with their frequency increasing by approximately 2.2-fold between 2003 and 2023 (Cunningham et al., 2024). These mega wildfire events have also grown more intense, generating their own weather systems and creating highly unpredictable behavior (TIME, 2025), additionally complicating suppression efforts and public communication.

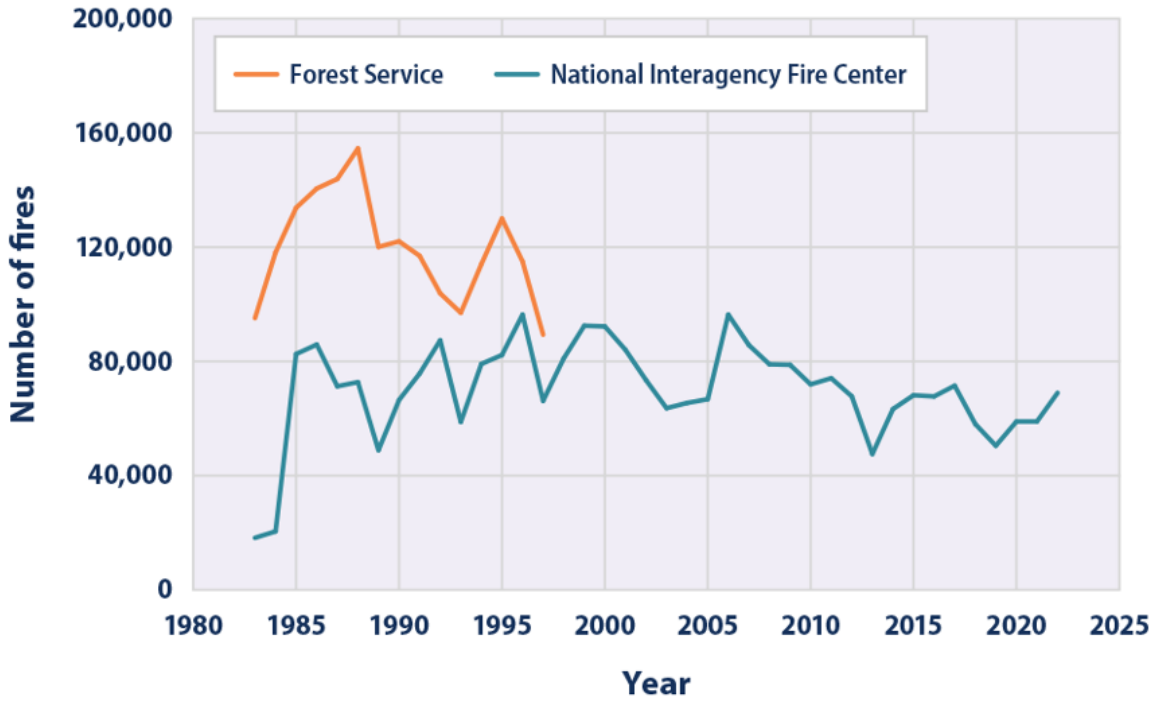


Figure 1 Number of wildfires in the United States per year reported by the Forest Service and the National Interagency Fire Center, 1983-2023. Source: United States Environmental Protection Agency (2022).

Wildfires are often overlooked as a serious problem until their scale and damage reach extreme proportions. While ecosystems in places like Australia have adapted to frequent fires during the normal evolution of the local fauna and flora, other areas such as California in the United States and the Pantanal biome in Brazil have faced a dramatic decline in ecological health since they are not adapted to face these events, driven by land degradation, climate change, altered land use and a growing frequency of wildfires (Figure 2). The 2024 fire season ranks among the most severe in the past two decades (Copernicus Atmosphere Monitoring Service, 2024), and predictions indicate this will not be the most severe for long.

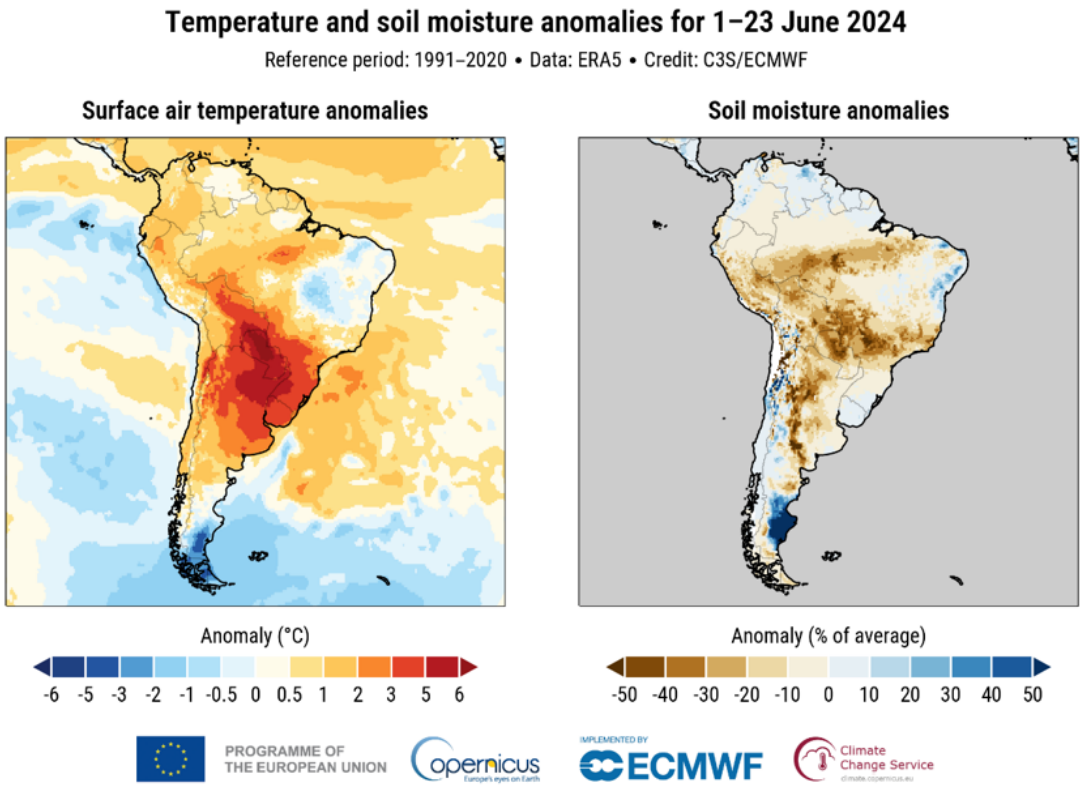


Figure 2 Temperature and soil moisture anomalies for 1-23 June 2024 in South America. Source: Copernicus Atmosphere Monitoring Service (2024).

To analyze and communicate wildfire behavior, geovisualization tools such as maps and WebGIS platforms are commonly used. However, traditional 2D interfaces frequently suffer limitations: users have restricted screen space and limited depth perception, making it

difficult to understand intricate temporal and spatial relationships. To deal with these issues, real-time 3D game engines provide dynamic camera angles which combine orthographic views with perspective flythroughs, enhancing spatial understanding and player involvement.

As severe wildfires become an annual occurrence, there is increasing demand for more effective methods to analyze and share wildfire data. This thesis aims to improve public and policymaker comprehension of wildfire impacts through immersive 3D visualizations, supplementing or replacing traditional charts, statistics, and 2D maps. Though numerical summaries and charts help with comparison, they often do not fully convey the spatial scale and visual impact of large wildfires.

Regardless of the benefits, 3D visualization tools also involve a lot of trade-offs, for example, high-fidelity scientific tools such as vFirelib provide detailed physical modeling of fire spread, including GPU-accelerated computation of fuel moisture effects and wind interactions. However, they commonly lack easy-to-use interfaces and cinematic polish, making them harder for non-expert stakeholders and general audiences to access (Wu et al., 2023). On the same page, web-based GIS dashboards typically offer static maps or limited 2D animations that, despite their data richness, fail to harness the narrative and visual communication strengths of modern 3D engines.

1.1 The Meaning of Modeling in this Thesis

Since the title of this thesis refers to wildfire spread modeling, it is important to state precisely the sense in which the term modeling is used, so that the scope of the work is not misread as the construction of a predictive physical fire spread model. A model is a structured abstraction of an observed phenomenon, and it need not be predictive to be a model. Building a mathematical or physical fire spread model, in which fuel moisture, slope, and wind are integrated into principal equations to forecast where a fire will advance, is one legitimate approach to the wildfire problem, and recent surveys map this wider landscape of detection, simulation, and prediction techniques under the framing of a wildfire digital twin (Huang et al., 2024). This thesis deliberately follows a different approach. The work presented here performs data-driven descriptive modeling and a spatiotemporal 3D reconstruction of satellite-derived fire observations, rather than a predictive physical simulation. The system does not forecast fire behavior, it reconstructs, organizes and represents what satellites have already

recorded, so that the extent, intensity and day-by-day progression of real events can be understood by non-specialist audiences.

The modeling in this thesis is therefore twofold. First, it is data modeling in the descriptive sense: raw MODIS detections are abstracted into clustered, temporally ordered and intensity-weighted entities that stand in for the underlying fire activity. Second, and equally important, it is visual modeling in the literal sense used in computer graphics. Within Unity, each fire is represented using particle systems, dynamic marker scaling based on FRP and detection density, and terrain meshes. Framing the contribution as a conscious choice to focus on 3D visual narrative and data representation, rather than on physical prediction, is both an accurate description of what was built and a deliberate scoping decision. It focuses on the communicative gap left by existing tools, which is the problem this thesis sets out to address.

1.2 Research Problem and Motivation

This thesis addresses a gap between the abundance of freely accessible fire observation data and the limited tools for communicating it to non-specialist audiences. NASA FIRMS provides near-real-time global fire detections at no cost, but its standard output is limited to two-dimensional maps and tabular downloads for fire management experts, which can be overwhelming for most of the audience that is not familiar with these data. Scientific simulators produce precise results but need specialized knowledge and hardware, making them unavailable to educators, journalists, and the public. There is a clear need for an open-source, reproducible pipeline that transforms observed fire data into an engaging, geographically accurate 3D visualization with a clear narrative which operates without proprietary dependencies and does not require programming skills.

1.3 Objectives

The aim of this thesis is to design and implement a modular Unity3D application that produces purely visual wildfire time-lapses without relying on any proprietary paid solution. The application is delivered in two builds. The desktop build is the reference implementation and the principal deliverable, containing the complete visualization described in Chapter 3. The browser build, produced for the WebGL platform, deploys the same pipeline without an installation step and is the build through which the application was made accessible for the comparative study reported in Chapter 5. The specific objectives are as follows:

- Develop a workflow that allows the proper visualization of wildfire extent and temporal progression using satellite-derived fire detection data from MODIS.
- Enable users to identify spatial and temporal patterns, gain a better understanding of real events and traverse an immersive 3D environment through a day-by-day timeline animation.
- Demonstrate that the complete pipeline can be implemented without any paid license or proprietary GIS platform, and that the workflow is reproducible and suitable for future fire events by practitioners without a programming background.
- Establish game engines, and Unity3D specifically, as a viable geovisualization platform capable of rendering georeferenced satellite data over accurate terrain, extending their use beyond interactive entertainment.

1.4 Contributions

This work furthers geovisualization and storytelling in geography by introducing a modular framework for building an immersive, three-dimensional application of observed wildfire events derived from geographical data. The framework needs no paid license and no dedicated server, and it connects game engines with professional geoinformatics through integrating .NET GIS libraries via NuGet, enabling runtime shapefile loading and precise geographic coordinate handling within Unity. It is delivered as two builds, a desktop reference implementation and a browser deployment of the same pipeline.

2. Background and Literature Review

To understand the significance of this thesis, it is important to understand the current state of wildfire visualization and simulation systems. This chapter reviews wildfire ecology not only as a destructive force, but also as an environmental phenomenon, assesses the primary satellite-based monitoring infrastructure, and explores the current range of existing tools. It compares development platforms to emphasize the advantages of game engines and narrative cartography for modern environmental communication. Together, these topics lay the scientific and technical basis for the Unity3D workflow discussed in later chapters. This work treats visualization not only as a method for displaying geospatial data but also as an essential means for increasing public and policymaker appreciation of complex spatio-temporal environmental patterns.

2.1 Wildfire Ecology and the Growth of Anthropogenic Threat

Wildfires naturally serve as important ecosystem regulators. Before analyzing how human activities have changed fire regimes globally, it's important to understand their role in a natural setting. Wildfires are not inherently destructive, rather, in many ecosystems, they are vital processes that help control fuel loads, promote seed germination, and sustain biodiversity cycles. Effective fire regimes depend on at least four factors: primary productivity, fuel structure, seasonality, and ignition sources (Keeley and Pausas, 2022). Additionally, fire regimes may differ considerably within the same ecosystem due to local variations in topography, vegetation density, and microclimate, resulting in indistinct burning patterns. The challenge arises when human pressures modify these regimes in ecosystems where fire was historically rare or absent. For example, in the Amazon, Barlow et al. (2020) link extreme fire activity directly to deforestation and poor environmental governance, noting that fires in August 2019 were three times more numerous than in August 2018. Across South America, projections indicate that the area affected by wildfires could increase by up to 32.7% by the late 21st century under current climate trends (Yang et al., 2026), leading to significant ecological changes in this and related ecosystems.

2.2 Satellite-Based Fire Monitoring: NASA FIRMS

Fire detection from space relies on the principle that burning vegetation emits strongly in the mid-infrared, making an active fire pixel unusually bright at around 4 micrometers

compared to its surroundings. Detection algorithms use this contrast to identify fire pixels and estimate the thermal energy released. NASA's Fire Information for Resource Management System (FIRMS) operationally implements this, originating in 2007 at the University of Maryland with NASA's and the UN's support, to provide timely fire location data as existing channels were too slow. It combines data from the Moderate Resolution Imaging Spectroradiometer (MODIS) on Terra and Aqua, offering a continuous record since 2000 at 1 km resolution, and the Visible Infrared Imaging Radiometer Suite (VIIRS) on Suomi NPP, NOAA-20, and NOAA-21, providing finer 375 m resolution but shorter archives. Detections are delivered within about three hours globally, faster in North America. However, the data presentation is limited; detections are accessed via browser maps and downloads designed for users familiar with fire pixels. Temporal behavior, essential to understanding fire spread, is only accessible through date filters, not continuous animation, and elevation data is missing by default. Users must reconstruct fire movement manually, highlighting a gap between data richness and its presentation, motivating the approach developed in the following chapters.

2.3 Existing Wildfire Visualization and Simulation Tools

Visualization tries to show data in a way that highlights patterns and facilitates understanding, without necessarily modeling the physical processes (Tirado Cortes et al., 2024). Simulation engines, by contrast, such as GPU-based fire spread models, compute physical interactions including fuel moisture dynamics and wind behavior to predict fire progression. While simulations provide scientific rigour, most of the time they lack good visual quality and intuitive interfaces that help non-expert audiences better understand complex spatio-temporal dynamics (Inglis and Vukomanovic, 2020). By avoiding reliance on complex physical models, the project becomes easier to develop and can employ simple visual tools to highlight the narrative message and improve public understanding (Evangelidis et al., 2018).

The actual landscape of wildfire visualization tools broadly divides into two categories: simulation-first tools that model fire-spread physics and visualization-first tools that render observed or historical fire data. Among simulation tools, vFirelib (Wu et al., 2023) represents a high-performance benchmark. vFirelib is a GPU-based fire simulation and visualization tool that uses a GPGPU (General Purpose Graphics Processing Unit) framework to parallelize fire-spread computations and achieve a 20x speedup over a sequential implementation of a widely used fire-spread model. To facilitate integration with other applications, the authors

implemented a wrapper including RESTful APIs to provide fire simulation as a service. In a related simulation-first effort, Meng et al. (2023) coupled a forest fire spread model with a dedicated visualization module to support fire extinguishing operations, illustrating the same paradigm applied to operational firefighting rather than public communication. More recently, immersive interfaces built specifically for professional responders, such as the virtual reality wildfire visualization system for firefighters proposed by Tirado Cortes et al. (2025), extend this simulation-first tradition into head-mounted display environments, but remain oriented toward trained operators rather than the general public audience that motivates this thesis.

Among visualization-first tools, the most common are web-based GIS dashboards such as those provided by FIRMS, GWIS, and EFFIS, which offer broad accessibility but constrain users to static or limited 2D animations. These platforms allow users to visualize and download filtered current and historical hotspots, create active fire alerts and access area statistics. They are data-rich but visually flat: they communicate fire locations without conveying the lived scale, temporal progression, or intensity that generates public understanding and urgency.

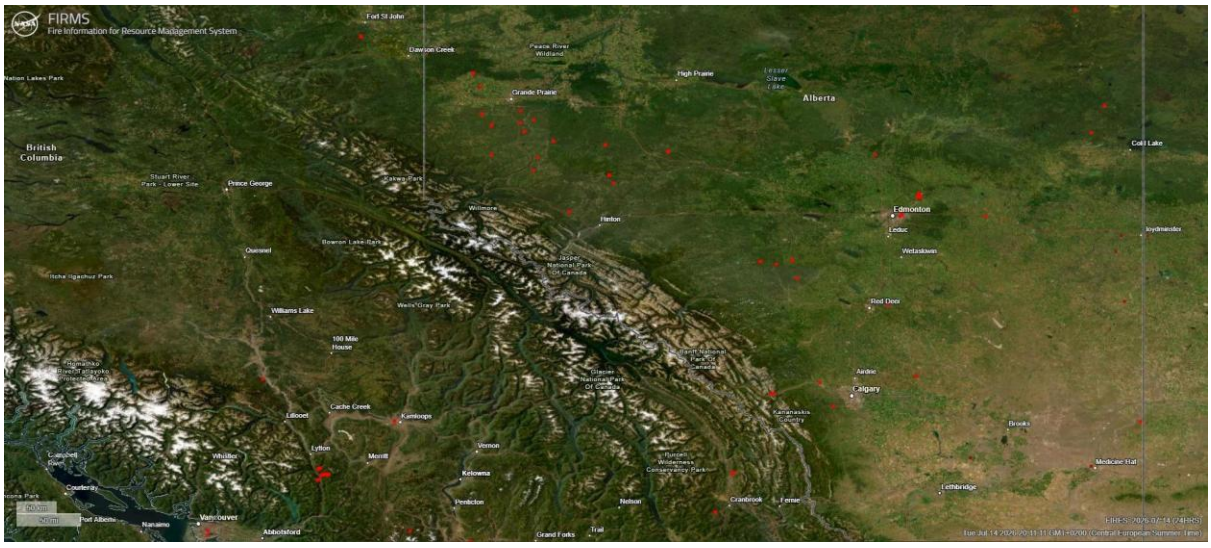


Figure 3 Global active fire detections displayed in the NASA FIRMS fire map. Source: NASA FIRMS (2026).

2.4 Platform Selection: Alternatives Considered

Before committing to Unity3D as the development environment for this visualization, four other platforms were evaluated: Unreal Engine 5, Blender with the BlenderGIS add-on, CesiumJS and ArcGIS Pro. Each one of them offers functional geospatial capabilities yet

presents trade-offs regarding accessibility, interactivity, real-time performance, and integration with open data pipelines, which ultimately led to Unity3D being selected.

Unreal Engine 5 is the most technically comparable alternative to Unity3D and in several respects it surpasses it in raw image quality. Its nanite-virtualized geometry system and Lumen global illumination pipeline produce photorealistic terrain and lighting that exceed what Unity's Universal Render Pipeline can achieve with equivalent hardware budgets. Game engines are powerful software systems that have surpassed their gaming origins and evolved into essential platforms for rendering and emulating complex virtual environments, and Unreal Engine 5 in particular applies advanced graphics rendering features, physics simulations and scripting systems to empower geomatic practitioners to construct arresting visual representations of spatial data (Pavelka and Landa, 2024). In the wildfire domain, research has demonstrated that Unreal Engine 5, combined with the AirSim simulator framework, can generate large-scale synthetic fire datasets across 16 km2 forest maps with controllable weather and fire-type parameters, confirming its suitability for high-resolution environmental scenarios (Sun et al., 2026). Although Unreal Engine's primary scripting model relies on C++ and a visual Blueprints system, for a data pipeline that requires custom C# integration of NetTopologySuite shapefile readers and Cesium coordinate bridges, the absence of a native C# environment would have required rebuilding significant portions of the data ingestion logic from scratch. In addition, Unreal Engine's larger binary size, higher hardware minimum requirements, and steeper learning curve for non-game developers make it harder to access in the academic and public-facing communication contexts this thesis targets.

Blender with BlenderGIS represents the open-source and artistic pole of the alternatives. Blender is primarily used to create images, animations, and 3D models, and it shines in situations where something more creative or visually distinctive is required, with animation capabilities that far exceed those of standard GIS tools in many respects. The BlenderGIS add-on allows direct import of GIS formats including shapefiles, GeoTIFFs and terrain models and Blender's Python scripting environment can automate batch processing of geospatial data. Nonetheless, Blender is basically a fundamental content creation and rendering tool rather than a real-time interactive engine. It does not support interactive runtime navigation of a 3D scene by an end user and the real-time viewport is not designed to handle continuously streaming terrain tile data or dynamically spawned particle systems responding to live data inputs. Any temporally animated product built in Blender would need to be exported as a pre-

rendered video file, losing the interactive timeline control, free-fly camera navigation and toggleable overlay layers that are fundamental to this thesis's contribution. For projects that require a fixed cinematic output rather than an interactive experience, Blender would be an excellent choice. Still, for this work, the lack of a true runtime environment is a disqualifying constraint.

CesiumJS occupies an interesting position because it shares the same underlying globe infrastructure as Cesium for Unity. CesiumJS is an open-source JavaScript library that creates 3D globes and maps, offering first-class support for time-dynamic simulation, live telemetry streaming, and 4D visualization (Cesium, 2025). Since Cesium operates entirely within the browser's JavaScript runtime, it has strict limits on the number of simultaneously active 3D primitives, particle effects and custom shader operations before frame rate can quickly become a problem . Some of the features and post-processing effects that constitute the visual design of this platform would demand considerable custom WebGL shader development to achieve in CesiumJS, work that is provided out-of-the-box by Unity's Particle System, VFX Graph and Post-Processing Stack.

ArcGIS Pro is the industry-standard professional GIS platform and offers genuine 3D scene capabilities through its Local and Global Scene view modes. It can ingest FIRMS shapefiles natively, apply temporal filtering via its time slider and render point features in 3D on elevation surfaces without any custom development. On the other hand, ArcGIS Pro is a commercial product with significant licensing costs, which directly undermine the open-access and reproducibility goals of this thesis. More importantly, traditional GIS environments leave many visual effects unutilized: things like particle-based fire effects, reflective surfaces and post-processing overlays, which are precisely what transform a geospatial dataset into a persuasive visual narrative (Hansen, 2021). ArcGIS Pro's 3D capabilities are oriented toward analytical accuracy and professional GIS workflows, they are not designed to produce the cinematic, narrative-driven, publicly accessible visualizations that are the central goal of this work.

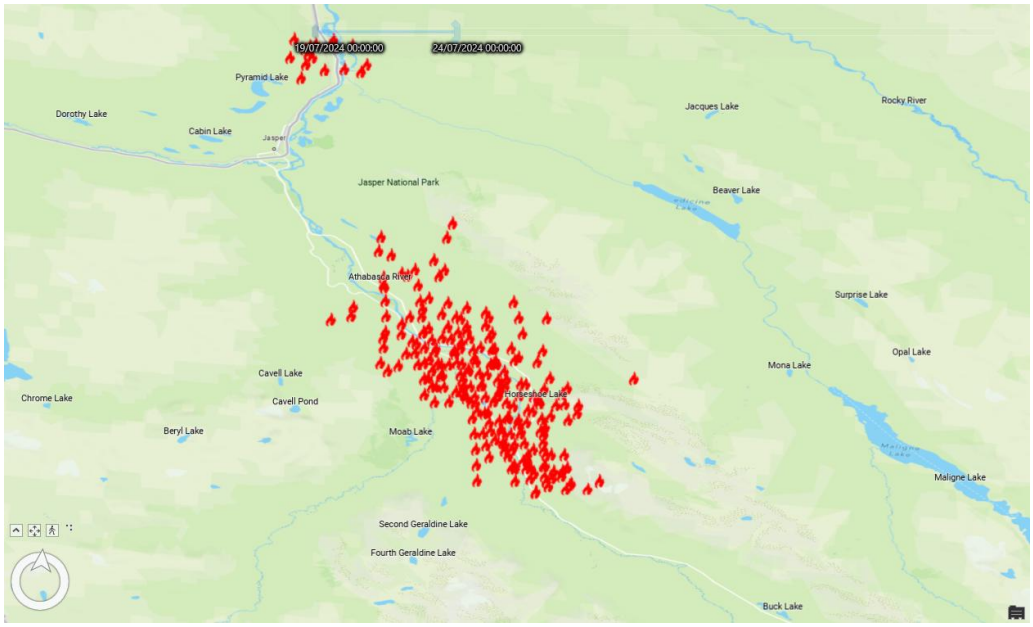


Figure 4 MODIS fire detections of the 2024 Jasper wildfire displayed in ArcGIS Pro. Source: author.

One final contender for the animations is Google Earth Pro. In terms of rendering speed, it falls behind all other evaluated tools (Figure 5), in particular for large volumes of data. However, Google Earth Pro offers a distinctive capability: creating presentation slides directly within the application, enabling structured storytelling alongside the visualization. Although Google Earth Pro was not used for the implementation, several functionalities of the application developed for this thesis were inspired by its storytelling capabilities.

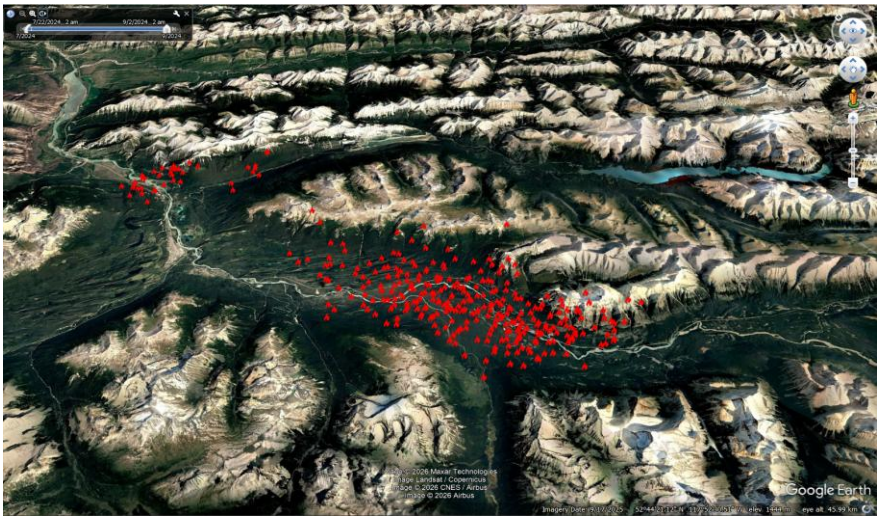


Figure 5 MODIS fire detections of the 2024 Jasper wildfire displayed in Google Earth Pro. Source: author.

Unity3D was selected over these four alternatives because it uniquely combines a C# scripting environment compatible with the NetTopologySuite geospatial library, native integration with Cesium for Unity's double-precision globe infrastructure, a production-grade particle and post-processing pipeline that can be driven by runtime data and a cross-platform deployment model that produces self-contained standalone applications needing no licenses, servers, or persistent internet connectivity. Table 1 summarises the key capability differences across all five platforms.

Table 1 Platform comparison across key capability dimensions

Criterion	**Unity3D**	**Unreal Engine 5**	**Blender + BlenderGIS**	**CesiumJS**	**ArcGIS Pro**
Real-time interactive navigation	Full	Full	None	Full	Limited
Native shapefile support	Via plugin	Via plugin	Full	Conversion needed	Full
Particle systems / VFX	Full	Full	Render only	Custom WebGL only	None
Temporal animation (timeline)	Full	Full	Pre-render only	CZML support	Time slider only
Cesium terrain integration	Full	Full	None	Native	None
Post-processing effects	Full (URP stack)	Full	Full (render only)	None	None

Criterion	Unity3D	Unreal Engine 5	Blender + BlenderGIS	CesiumJS	ArcGIS Pro
Free to use	Free tier	Free tier	Open source	Open source	Paid license
Standalone executable	Yes	Yes	No	Browser + server	License required
C# scripting	Yes	C++ / Blueprints	Python	JavaScript	Python / ArcPy
Cross-platform deployment	Yes	Yes	Yes	Any browser	Windows only
VR / AR support	Yes	Yes	Limited	Limited	None
No server / internet required	Yes	Yes	Yes	No	Partial
Learning curve	Low to medium	High	Medium	Medium	Medium

2.5 Game Engines for Geospatial Visualization

Game engines such as Unity3D and Unreal Engine are increasingly recognized as platforms for geospatial visualization that extend well beyond their traditional entertainment domain. Although originally designed for interactive entertainment, they can also be applied in scientific and training contexts (Trenholme & Smith, 2008), a potential increasingly reinforced by systematic evidence that gamified approaches improve wildfire education and safety training outcomes (Huang et al., 2025). Their real-time rendering capabilities offer

unique affordances for large-scale, dynamically evolving environmental scenes that exceed the performance limits of traditional desktop GIS software. Together, these capabilities enable rapid development of polished, narrative-driven visualizations without demanding extensive low-level graphics programming. GIS-to-game-engine workflows have underscored these capabilities as key enablers of real-time spatial data storytelling, and game engines are known to deliver a deeper, immersive, real-world-like experience using actual geodata (Laksono & Aditya, 2019). Buyuksalih et al. (2017) presented a detailed Unity-based 3D urban modeling case study documenting the full workflow from GIS mesh import to texturing, LOD configuration, and performance optimization. Their work underscores Unity's asset management ecosystem as a substantial advantage over traditional GIS software for cooperative and iterative development, while noting restrictions in processing very large shapefile datasets natively within the editor. Laksono and Aditya demonstrated in 2019 that Unity3D can integrate mixed GIS sources, including terrestrial laser scanner models and topographic map data, into multi-platform 3D visualizations that display LOD3 models and provide user interfaces for on-the-ground and aerial first-person exploration. Unity's Timeline system further abstracts animation sequencing, allowing non-programmers to keyframe object visibility and effect parameters via a drag-and-drop UI (Unity Technologies, 2025).

2.6 Narrative Cartography and Immersive Spatio-temporal Storytelling

Effective wildfire communication depends not only on technical rendering capacity but on how spatial and temporal data are narrated. Good communication of environmental phenomena hinges crucially on narrative structure. Caquard and Cartwright (2014) define narrative cartography as the discipline that studies the relationship between the nature of stories and maps, arguing that maps are not only neutral data containers but inherently narrative artifacts, and that making such narrative structure explicit produces measurable gains in comprehension. Storytelling is a technique for conveying the messages embedded in data to a broad audience. It is well suited for geographic and GIS applications, particularly when interactivity and 3D visualizations are involved, as immersion into the story and the visualized data is thereby facilitated (Caquard & Cartwright, 2014; Cartwright & Field, 2015).

Thöny et al. (2018) identify three core mechanisms that work particularly effectively for 3D geographic storytelling. First, temporal sequencing: guiding viewers through events in chronological order using visual cues such as fade transitions and particle flow. Second, focal

emphasis, which is the use of visual effects such as bloom or glow to highlight key elements, which in the context of this thesis means advancing fire fronts across several scenarios. Third, contextual framing: embedding dynamic event layers within static terrain and land-cover contexts to situate the phenomenon in its actual-world environment. In contemporary cartography, storytelling is mainly achieved through story maps; however, these story maps largely neglect the narrative character of the map itself, resulting in interactive maps that present independent thematic features or spatially referenced individual stories rather than map-centered, consistent narratives. The Unity-based approach proposed here precisely tackles this limitation by embedding the narrative directly into the scene's spatial and temporal structure.

3. Methodology

This chapter describes the complete methodological structure underpinning the wildfire visualization platform. The methodology tackles three interconnected concerns: how geospatial and meteorological data are acquired and pre-processed, how those data are integrated into the Unity3D WebGL rendering environment via a coordinate system bridge that resolves floating-point precision limitations at geographic scales, and how the resulting fire, terrain and weather layers are assembled into an interactive spatio-temporal narrative.

In preparing this thesis, generative artificial intelligence tools were used as a writing and development aid. Their use was limited to language editing and restructuring of text drafted by the author, assistance with writing and debugging code for the application and the data acquisition pipeline, and assistance with statistical analysis. All research questions, study design, data collection, interpretation of results, and conclusions are my own. Every statement, figure, and result in this thesis has been verified by me, and I take full responsibility for its content.

3.1 Workflow Overview

The methodology follows a three-phase pipeline designed to be entirely self-contained within the Unity Editor environment, requiring no external GIS software, no proprietary runtime licenses, and no manual intermediate data processing. The three phases are: (1) data acquisition, in which fire detection records are retrieved from the NASA FIRMS API, weather data are fetched from Open-Meteo, and all outputs are exported as open-standard geospatial files; (2) data processing and Unity integration, in which those files are loaded into Unity, geographic coordinates are converted to Unity world positions using Cesium for Unity's coordinate transformation infrastructure, fire detections are clustered for performance, and animated 3D markers are spawned on the photorealistic globe; and (3) interactive visualization, in which the user navigates a real-time 3D environment, controls a day-by-day temporal timeline, and explores spatially contextualized fire, wind, and temperature overlays simultaneously. The phases and their steps are shown in Figure 6.

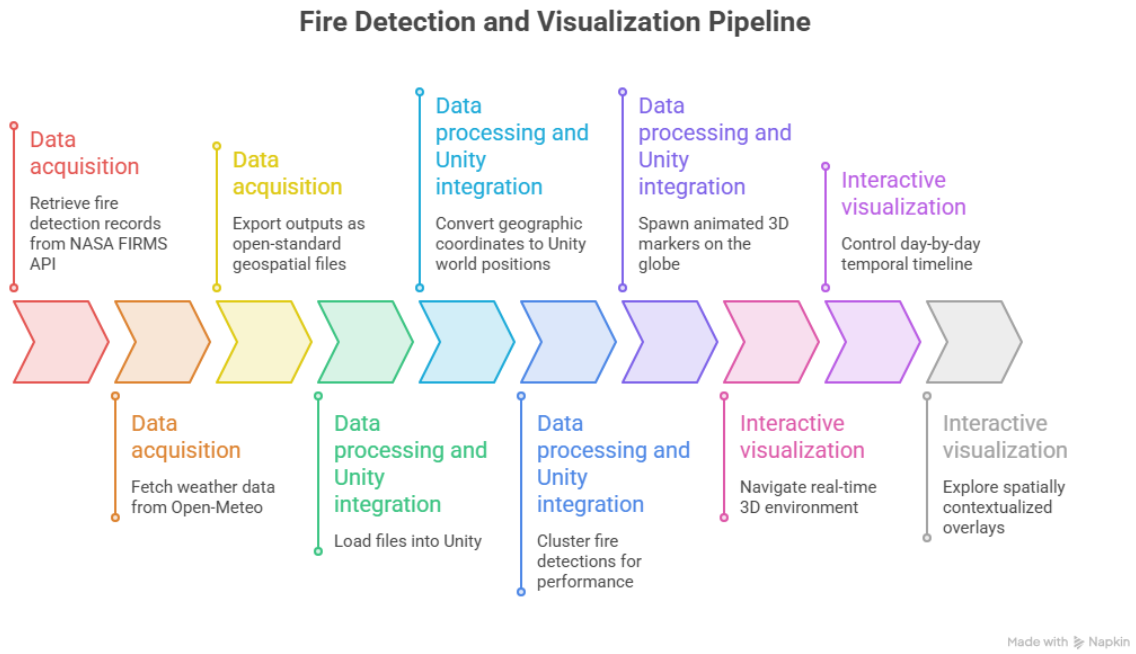


Figure 6 Overview of the fire detection and visualization pipeline, from data acquisition through data processing and Unity integration to interactive visualization. Source: author.

The core technical challenge around all the phases is the bridging of two fundamentally incompatible coordinate systems: the WGS84 geographic coordinate system (latitude/longitude in decimal degrees) used by all earth observation datasets and the local Cartesian coordinate system (floating-point meters relative to an arbitrary origin) used by the Unity renderer. Every object placed in a Unity scene has a position defined by three floating-point numbers, X (horizontal), Y (vertical) and Z (depth), measured in meters from a fixed reference point called the world origin at (0, 0, 0).

When real-world geographic data is introduced, a problem arises in aligning it with the Unity landscape. Geographic coordinates (latitude, longitude, and elevation) are measured relative to the center of the Earth, not to a convenient point on its surface. Unity's internal arithmetic uses single-precision floating-point numbers to represent positions, and at distances of millions of meters from the origin, these numbers lose precision: the smallest representable step is no longer millimeters but entire meters, or even tens of meters. In practice, this produces visible jitter on moving objects, fire markers that drift from their correct geographic location as the camera moves, and terrain geometry that snaps rather than flows smoothly.

Cesium for Unity solves this by relocating the coordinate origin to the geographic centroid of the loaded fire dataset, so that every object in the scene is at most a few hundred kilometers from the origin rather than millions. It also uses double-precision arithmetic internally and automatically recalculates positions as the camera moves across the globe using a mechanism called origin shifting. The result is that every fire marker, terrain tile and weather arrow can be placed at its correct geographic location with sub-metre accuracy, regardless of the study area size or the camera position.

3.2 Study Area: Jasper National Park, Alberta, Canada (2024)

Bounding box: 119.0°W, 52.5°N, 117.0°W, 53.5°N | Area: ~11,000 km² (Jasper NP core)

The 2024 Jasper wildfire was selected for this scenario because of its urban-wildland interface fire dynamics within a compact, high-elevation terrain. The 2024 Jasper wildfire was a complex of fires that burned in Jasper National Park, Alberta, Canada, in July 2024. Fires ignited north and south of the resort town of Jasper and quickly grew out of control, forcing the mass evacuation of 25,000 residents, workers and visitors. The fires merged and swept through the town, destroying 358 of its 1,113 structures. Insurance companies paid $880 million in claims, making it one of the most expensive natural disasters in Canadian history. The final fire size was 32,722 hectares (Parks Canada, 2024).

From the first ignition to the fire reaching the townsite was less than 48 hours. During those first two days, under hot, dry and windy conditions, flame heights reached 30 to 50 meters, with new spot fires starting up to 500 meters ahead of the fire front from lofted embers. The Canadian Forest Service confirmed lightning as the ignition source for the fires south of town. This creates a short but extremely intense FIRMS detection sequence, a few days of very high FRP values concentrated in a small geographic area, which tests the platform's ability to convey dramatic temporal acceleration in the timeline animation against a relatively sparse overall detection dataset.

3.3 Input Data

The input data used by this platform comes from three distinct open sources: NASA FIRMS fire detection records, Open-Meteo historical meteorological records and Cesium World Terrain tiles. This section describes each source in detail, including its spatial and

temporal resolution, the format in which the pipeline consumes it and the specific limitations that constrain its use.

3.3.1 Fire Detection Data

The pipeline consumes the MODIS Collection 6.1 active fire product distributed by NASA FIRMS, described in Section 2.2. MODIS was selected over VIIRS for three reasons. Its Terra and Aqua platforms together yield up to four overpasses per day, providing denser temporal sampling than a single-platform sensor. Its archive extends unbroken to 2000, which permits historical events to be reconstructed on the same basis as recent ones. And it reports confidence as a numerical value from zero to 100 rather than as a categorical class, which allows a threshold to be set and varied precisely, a property the filtering described in Section 3.5 depends on. The coarser one-kilometer pixel is a real cost of that choice, but at the regional extents visualized here it does not affect the legibility of the reconstruction, and a single consistent data model across all scenarios simplifies both the pipeline and the cross-tool comparison in Chapter 4.

3.3.2 Meteorological Data

To contextualize fire behavior within the atmospheric conditions that drive it, the platform integrates historical weather data from the Open-Meteo Historical Archive API. Daily records are retrieved on a spatial grid covering the fire bounding box, with configurable point spacing. Each record includes the daily mean, maximum, and minimum air temperatures at 2 m above ground; wind speed in meters per second and wind direction in degrees at 10 m; and mean and maximum cloud cover percentages. All weather records are stored as JSON files in the Unity project's StreamingAssets/WeatherCache directory and loaded at runtime by the visualization system.

3.3.3 Terrain and Imagery

The three-dimensional surface on which all fire and weather data are rendered is provided by Cesium for Unity, which streams the Cesium World Terrain dataset, based on Airbus WorldDEM4 Ortho, as quantized mesh tiles from the Cesium ion cloud platform. Sentinel-2 cloudless imagery from EOX is draped over the terrain tiles, providing photorealistic satellite textures without any pre-download or local storage requirement. Tiles are loaded on

demand, scaled by camera distance and a screen-space error threshold, making the system responsive at both continental overview altitudes and ground-level perspectives.

3.4 Cesium Coordinate System

A structural incompatibility exists between the two coordinate systems this platform must reconcile: the WGS84 geographic coordinate system used by all Earth observation datasets, expressed as latitude and longitude in decimal degrees, and Unity's local Cartesian coordinate system, expressed as floating point meters measured from an arbitrary origin. Every object placed in a Unity scene occupies a position defined by three floating point numbers, X (horizontal), Y (vertical) and Z (depth), measured from a fixed reference point at (zero, zero, zero). Geographic coordinates, by contrast, are measured relative to the center of the Earth rather than a convenient point on its surface, so an unmodified fire marker placed on the surface of Australia would sit several million meters from that origin along all three axes at once.

This distance matters because Unity's internal arithmetic uses single precision floating point numbers to represent positions. At distances of millions of meters from the origin, the smallest representable step is no longer millimeters but entire meters, or tens of meters. Left uncorrected, this produces visible jitter on moving objects, fire markers that drift from their true geographic location as the camera moves, and terrain geometry that snaps rather than flows smoothly, all of which would directly undermine the objective of representing fire location and progression with geographic fidelity.

Cesium for Unity resolves this by relocating the coordinate origin to the geographic centroid of the loaded fire dataset at application startup, so that every object in the scene sits at most a few hundred kilometers from the origin rather than millions. Internally, Cesium performs all coordinate transformations in double precision, converting between the WGS84 ellipsoid, an Earth Centered, Earth Fixed representation and Unity's local space (Cesium, 2025). As the user's viewpoint moves across the globe, an origin-shifting mechanism recalculates every object's local position relative to a nearby new origin, so that vertex precision near the camera is preserved regardless of how far the camera has traveled from the scene's starting point. Because every fire marker retains its underlying Earth Centered, Earth Fixed coordinate rather than only its local Unity position, this recalculation happens without drift accumulating over time. The result is that fire markers, terrain tiles and weather overlays

remain correctly placed to sub metre geographic accuracy across study areas of very different sizes.

A second consequence of this architecture is that terrain and fire data load asynchronously. Cesium streams terrain tiles progressively rather than all at once, so terrain geometry beneath a given fire location may not yet be present while that location's marker would otherwise be instantiated. Spawning a marker before its terrain has loaded would place it visibly above or below the ground or prevent it from attaching to a surface at all. To avoid this, the platform verifies that terrain is present beneath a representative sample of fire locations before allowing markers in that region to spawn, briefly retrying where necessary.

3.5 Data Acquisition Pipeline

The data acquisition pipeline is constrained by a limitation inherent to the FIRMS API: fire detections must be queried five days at a time rather than as a single range request. The pipeline accommodates this by iterating across the full user-specified date range and concatenating the daily results into a single dataset. Because Earth observation downloads of this kind are routinely interrupted by transient network failures rather than by errors in the request itself, the acquisition step retries automatically rather than requiring the user to restart a multi-day download from the beginning. The final dataset is exported as both a CSV file and a point shapefile in EPSG:4326, carrying the standard FIRMS attributes described in Section 3.3.1, so that the same output can be consumed by the Unity pipeline or inspected independently in conventional GIS software.

Confidence filtering is applied at two distinct stages rather than once, and the two stages serve different purposes. At acquisition, the exported dataset retains detections at nominal confidence and above, consistent with the minimum established in the peer-reviewed literature for fire extent studies (Giglio et al., 2016). This preserves the full extent of the recorded event in the archived dataset, including the lower-radiative-power activity that a stricter cutoff would remove, along with the noise it is intended to suppress.

At runtime, the data loader applies a second, stricter threshold, retaining only detections in the high-confidence band. This threshold responds to two constraints. The first is computational: each retained detection instantiates a particle system, a smoke emitter, and a burn-scar mesh, so the number of detections carried into the scene directly limits what can be

rendered at an interactive frame rate, and this constraint binds most tightly in the browser deployment. The second is communicative. The platform addresses an audience without the domain knowledge to discount a provisional detection, and a marker rendered as fire is read by such an audience as fire. Restricting the scene to high-confidence detections therefore reduces the risk of presenting probable false positives to viewers who cannot evaluate them, a risk an analytical tool addressing specialists would not need to manage.

The cost of this decision is a systematic under-representation of fire extent, concentrated at the margins of the front and in the opening and closing hours of a burn, where radiative power and confidence are both lowest. That cost is quantified in Section 5.5. The threshold is exposed as a configurable parameter rather than fixed in code, so a user prioritising completeness over rendering budget can lower it toward the acquisition minimum.

Meteorological context is retrieved from the Open Meteo Historical Archive API, which, unlike the FIRMS product, returns data for individual coordinate points rather than for an arbitrary polygon. This point based access model means weather cannot simply be requested for the fire's bounding box, sampling locations were instead distributed across the extent on a regular grid, a design chosen to give the platform a spatially varying picture of wind and temperature conditions, consistent with the wind sway.

At runtime, the fire shapefile is read through NetTopologySuite, the same library used throughout the pipeline and parsed into a structured list of fire records. Locating the correct output file automatically, rather than requiring the user to specify a path by hand each time the pipeline is rerun, was a deliberate usability decision aimed at the third objective, a workflow operable without a programming background.

3.6 Temporal Animation

The temporal animation system is built around a day-by-day playback structure rather than a continuous or free scrubbing timeline, because the second objective set out in Section 1.3 is for users to follow the spatial and temporal progression of an event through discrete daily observations (Figure 7), matching the way satellite detections are themselves delivered as one pass per day rather than as continuous coverage. Detections are organized chronologically, and each daily transition fades out the previous day's markers while fading in the new day's, so the

viewer's attention is drawn to change over time rather than to the mechanical appearance and disappearance of individual points.

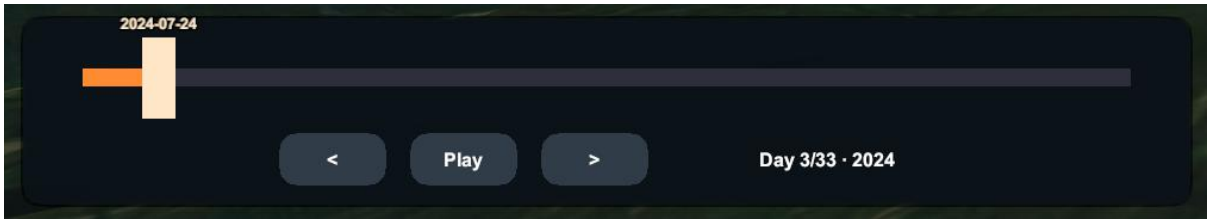


Figure 7 Timeline control of the Unity application. Source: author.

In the desktop application, fire markers are scaled using a formula that combines the base marker size, an editable size multiplier, the square root of the detection's fire radiative power normalized to the dataset's median, and the square root of the number of detections clustered around that point. Both normalization terms were deliberately compressed with a square root rather than scaled linearly because fire radiative power in raw satellite data spans several orders of magnitude, from a small smoldering detection to an extreme, high-intensity fire front. A linear scale would cause the small number of most intense detections to visually dominate the scene, making moderate-intensity fire activity imperceptible by comparison.

3.7 Weather Integration and Visualization Overlays

Wind data loaded from the Open-Meteo JSON cache is applied to the scene in two concurrent ways. First, each fire marker's sway animation parameters, the direction and amplitude of the particle system's lateral motion, are updated on each timeline day to reflect the wind vector at the nearest weather grid point, producing a spatially varying sway field in which fires in different parts of the scene respond to locally different wind conditions (Figure 8) . Second, a wind arrow grid is instantiated as an orthographic array of arrow GameObjects over the fire extent, with each arrow's rotation set to the local wind direction and its visual scale proportional to wind speed.

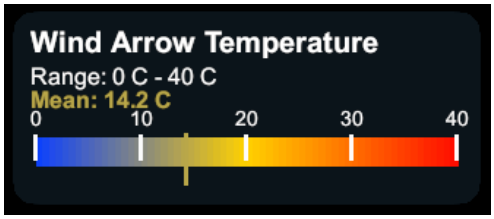


Figure 8 Wind arrow temperature legend of the Unity application. Source: author.

3.8 User Interface and Interaction Design

The interaction design follows directly from the three storytelling mechanisms identified in Section 2.6, temporal sequencing, focal emphasis and contextual framing, translating each into a specific interface element rather than leaving the narrative structure implicit in the scene alone. Temporal sequencing is exposed to the user through an explicit timeline control showing the current date against the total day count, allowing the day by day progression established in Section 3.6 to be navigated directly rather than only observed passively. Contextual framing is supported by concurrent weather readouts, cloud cover and wind speed and direction for the day being viewed, so fire activity is never presented in isolation from the atmospheric conditions argued in Section 3.7 to drive it. Overlay controls allow secondary layers such as the wind arrow field to be toggled independently of the fire markers themselves, a deliberate choice to let the user isolate the phenomenon of interest, fire spread against wind context, rather than presenting a fixed combination of layers the user cannot adjust.

Free navigation was chosen over a fixed or scripted camera path because the second objective of the thesis is for users to identify spatial and temporal patterns for themselves, which requires the ability to move between a continental overview, where the overall extent and shape of the fire event is visible, and a ground level perspective, where individual clusters and the surrounding terrain can be inspected closely. A scripted camera could guarantee a consistent viewing experience across users, but at the cost of removing exactly the exploratory freedom the objective calls for; free navigation was judged the better fit, despite the added onboarding cost that Chapter 6 later identifies as a source of the usability gap for first-time three-dimensional users. This same interaction model, timeline control, weather context and free camera navigation, is what participants were asked to use during the guided task described in Section 4.5, to locate the affected area, follow the fire across days and identify the day of greatest activity.

3.9 Unity Development

The framework is delivered in two builds rather than one because the deliverable and the instrument used to evaluate it place incompatible demands on the same project. The

deliverable must achieve the highest visual and geographic fidelity the pipeline can provide. The evaluation, by contrast, requires a build that a nonspecialist participant can open in a browser alongside two browser-based comparators, since any installation step would have removed the comparability on which the study design in Section 4.3 depends. A desktop application and a browser deployment were therefore built to serve each requirement separately rather than compromising a single build between them.

3.9.1 Desktop deployment

The desktop application is the reference implementation of the framework, in which every visualization layer, animated fire markers, accumulated burn scars (Figure 9), weather driven wind arrows, a cartographic scale bar, and a minimap (Figure 10)to help the user locate where are they looking at, was developed and validated first. It is distributed as a self-contained Windows executable, requiring no installation, license or server connection beyond terrain streaming itself, a deliberate advantage sought over ArcGIS Pro and CesiumJS, neither of which can produce a single distributable, license free desktop application (Section 2.4). Execution begins in a main menu where the fire point data, the weather and cloud records, an optional georeference origin and the visual quality preset are configured before the visualization scene loads, so the same executable can be reused across different study areas without recompiling the project (Figure 11). A saved configuration can be stored and reloaded, allowing a prepared scenario to be reopened directly.



Figure 9 Accumulated burn scars in the desktop build. Source: author.



Figure 10 Minimap and compass in the desktop build. Source: author.

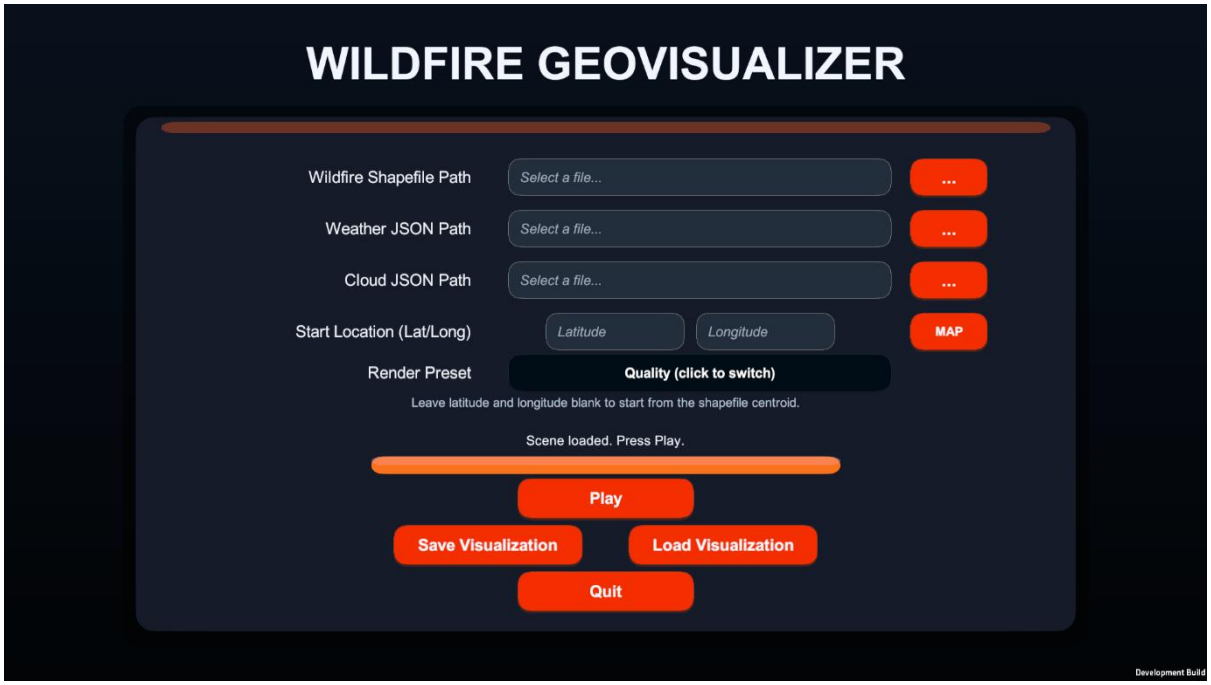


Figure 11 Launch configuration menu of the desktop build. Source: author.

This reference status matters methodologically as much as technically: because every feature exists first and most completely on the desktop build, it represents the upper bound of what the pipeline can communicate, against which the WebGL build's necessary compromises can be measured. The desktop application therefore provides the highest visual fidelity and the complete feature set of the framework, at the cost of requiring a Windows machine and dedicated graphics hardware that a member of the public is not guaranteed to have (Figure 12).

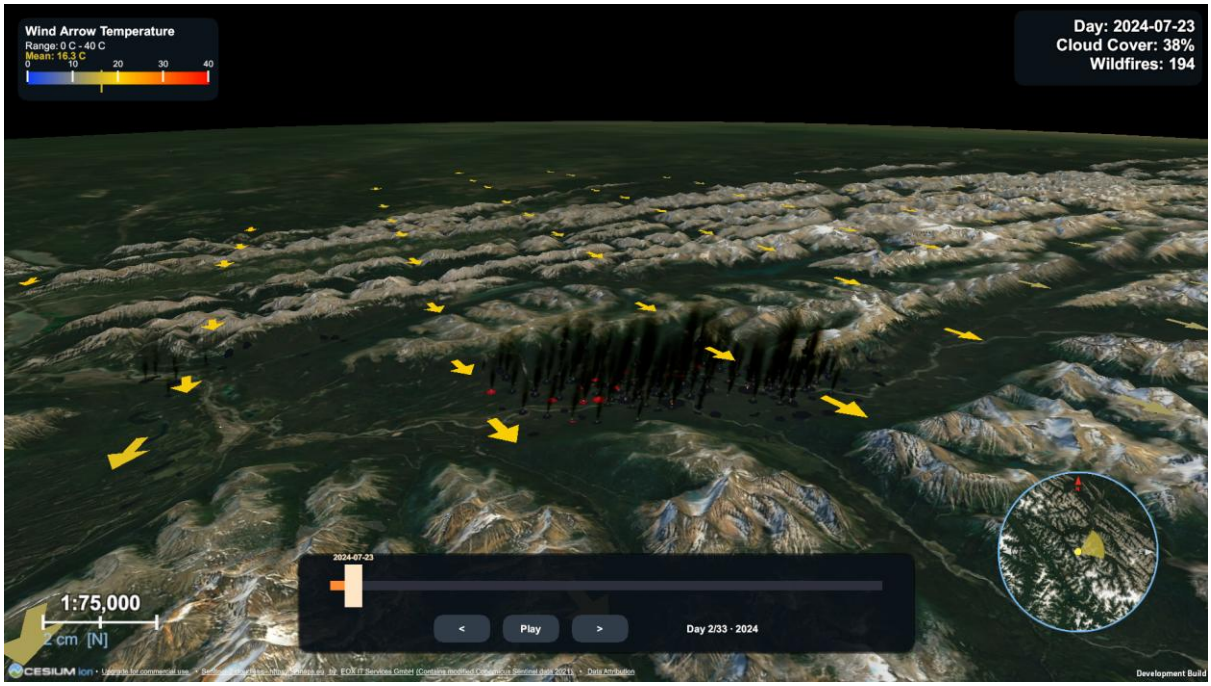


Figure 12 Desktop build showing the Jasper wildfire on 23 July 2024. Source: author.

3.9.2 WebGL Deployment and Browser Execution

The WebGL build exists specifically to remove that hardware and installation barrier, allowing a wildfire reconstruction, in the implementation evaluated in Chapter 4, the 2024 Jasper wildfire, to be explored directly in a web browser with no installation step at all. This accessibility is not free: because a browser cannot access the local file system directly and imposes stricter limits on the number of simultaneously active particle effects and geometry than a native executable, the WebGL build necessarily sacrifices some of the desktop version's visual density. The two builds are therefore best understood as a deliverable and the instrument through which it was evaluated, rather than as competing products. The desktop build carries the framework at full capability, and the browser build demonstrates that the same underlying pipeline, coordinate handling, clustering, and timeline animation, remains legible once the constraints of a browser runtime are imposed on it. What those constraints remove is reported in Section 5.5.

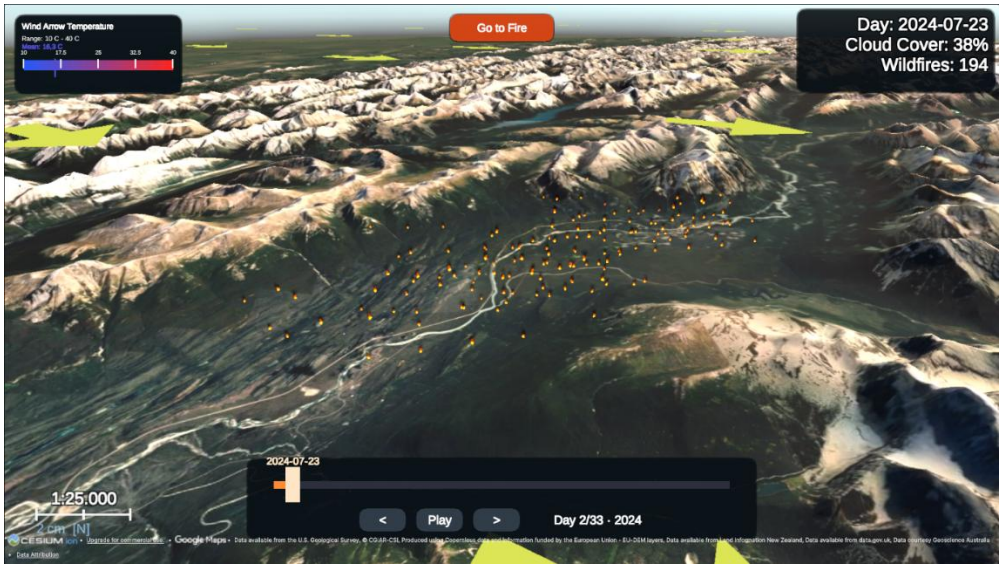


Figure 13 Browser build showing the Jasper wildfire on 23 July 2024. Source: author.

4. User Testing

This chapter describes the comparative user-testing study conducted to evaluate the communicative effectiveness of the visualization developed in this thesis. It states the aims of the study, describes the participants, the three tools and scenarios, the questionnaire instrument, the procedure and online delivery, the data analysis plan, and the design limitations. The outcomes of the study are reported separately in Chapter 5 and interpreted in Chapter 6.

4.1 Aims of the User Testing

The study addresses a single question: does the interactive three-dimensional reconstruction developed in this thesis communicate a wildfire event more effectively to a non specialist audience than two established tools, ArcGIS Online in the form of an Experience Builder application and kepler.gl, when all three present different data in order to remove the learning effect described in Section 4.4. This question follows directly from the research problem stated in Section 1.2: NASA FIRMS and its associated dashboards deliver rich fire detection data through two dimensional maps aimed at fire management professionals, while the pipeline developed in this thesis proposes an alternative aimed at non specialist audiences, and up to this point that alternative has only been argued for on design grounds, through the storytelling mechanisms of Thöny et al. (2018), the spatial cognition literature reviewed in Section 2.5 and the coordinate and rendering pipeline documented in Chapter 3. The user testing study exists to subject that design argument to an empirical test rather than leave it as an assertion.

The two tools chosen for comparison were not arbitrary. They are direct descendants of the platform elimination process carried out in Section 2.4: kepler.gl instantiates the browser based, terrain free style of visualization that Section 2.3 identifies as data rich but visually flat while the ArcGIS Online Experience Builder instantiates the professional, terrain aware GIS dashboard against which Section 2.4 measured Unity itself. Comparing the thesis platform against both, rather than against only one, allows the study to separate two questions that would otherwise be confounded: whether terrain matters at all, tested by the contrast with kepler.gl, and whether a game engine's cinematic and narrative apparatus adds anything once terrain is already present, tested by the contrast with the Experience Builder. The study is guided by five specific questions that follow from this structure:

1. Do participants understand the event correctly, that is can they answer factual questions about when, where and how the wildfire developed, and does the accuracy of those answers differ across the three tools?
2. How usable is each tool, as measured by an adapted usability questionnaire based on the System Usability Scale, and does usability differ across the three tools, which is necessary because a tool that communicates well but cannot be operated without assistance would not meet the accessibility aim stated in Section 1.2.
3. How well does each tool convey the spatial context of the event, that is the relationship between the fire and the surrounding terrain, and how well does each tool sustain the participant's attention, which tests the terrain and engagement claims made in Section 2.5 and Section 2.6 directly rather than by argument alone.
4. For the concrete elements of the visualization, namely the identification of the fire location, the legibility of the day by day development, the interpretability of the fire markers and their intensity, and finally the distinction between the recorded detections and the added visual effects, which are handled well and which poorly by each tool, since the second objective in Section 1.3 depends on all four of these being legible rather than on the platform succeeding at the visualization only as a whole.
5. Lastly, which tool is most effective for understanding where the event occurred and how it developed, and which is most suitable for explaining a wildfire event to a general audience, which is the question the thesis ultimately needs answered to claim the fourth objective in Section 1.3, establishing the platform as a viable alternative rather than only a technically functional one.

Comprehension, usability and engagement are kept as separate constructs throughout the instrument described in Section 4.4 precisely because collapsing them into a single satisfaction score would not allow a result to be attributed to the correct mechanism. A tool could, in principle, be enjoyable to use while communicating the event poorly, or could communicate the event correctly while being difficult to operate, and only a study that measures the three separately can distinguish between these outcomes.

4.2 Participants

In total, 68 participants took part in the study. Because the visualization is intended for a mostly non specialist audience, educators, policymakers and members of the public exposed

to wildfire coverage, rather than for fire management professionals who already have access to FIRMS and the GIS dashboards, participants were recruited from a general adult population and were not required to have prior experience with geographic information systems or with three dimensional applications. Restricting recruitment to a specialist population would have tested the platform against the wrong audience, since it is previously identified in this thesis that the gap is lying between rich data and non specialist communication, not between competing professional tools.

A short background section, answered once at the start of the questionnaire and described in full in Section 4.4, recorded four variables for each participant: age group and field of study or professional background, intended to describe how representative the sample is of a general audience rather than of a single discipline, and self rated experience with geographic information systems and with three dimensional or game environments, intended specifically to test whether any communicative advantage found for the Unity application in Section 4.6 depends on prior exposure to similar tools or is available to genuine novices, since only the latter would support the accessibility claim made in Section 1.3. A fifth variable, self rated familiarity with wildfire or satellite data, was also recorded to check that the comprehension results reported in Chapter 5 are not simply reproducing prior subject knowledge. Recruitment did not exclude participants with a technical background, and all five variables were treated as descriptive covariates for the analysis described in Section 4.6 rather than as a screening criterion, since excluding technically experienced participants would have removed exactly the comparison, novice against experienced, that the second background variable was collected to support.

A realised sample of 68 divides into three groups of comparable size, 21 participants who began with kepler.gl, 24 who began with the ArcGIS Online Experience Builder, and 23 who began with the Unity application, once the counterbalancing scheme described in Section 4.5 is applied. This size is sufficient to let the comprehension comparison in Chapter 5 rest on three independent groups large enough to show a clearer pattern where one exists, while remaining explicitly too modest to support formally generalisable statistical claims. It is, throughout, a convenience sample rather than a probability sample of any defined population, a limitation stated in full in Section 4.7 and carried forward into the interpretation in Chapter 6.

4.3 Materials: Three Tools and the Three Scenarios

Three tools are compared: the ArcGIS Online Experience Builder, kepler.gl, and the browser deployment of the application developed in this thesis. The browser deployment is used in place of the desktop reference build because all three conditions had to be reachable through a link, without installation, for the online delivery described in Section 4.5 to be possible at all. Each tool is paired with a different wildfire event: the Unity application shows the 2024 Jasper wildfire in Canada, the Experience Builder shows the Serra da Estrela fire in central Portugal in August 2022, and kepler.gl shows the Sierra de la Culebra fire in Spain in June 2022.

4.3.1 Selection of the Comparison Tools

The two comparators were chosen for the same reason the Unity implementation itself was chosen in Section 2.4, to test the platform against alternatives that a non specialist audience could plausibly reach today rather than against a hypothetical ideal tool. Before settling on kepler.gl and the ArcGIS Online Experience Builder, Google Earth Pro, Google Earth Web and Cesium ion were each considered and set aside for reasons specific to their fit as a comparator rather than as a development platform. Google Earth Pro was set aside because it requires local installation, which would have introduced a barrier to participation absent from the two web based tools and from the Unity WebGL build itself, undermining the comparability of the three conditions. Google Earth Web was set aside because it lacks a native timeline control comparable to the day by day progression central to the second objective in Section 1.3, which would have made any comprehension difference difficult to attribute to the tool's visualization rather than to the absence of a feature the other two tools possess. Cesium ion was set aside for a more specific reason, it shares the same underlying terrain streaming engine as the Cesium for Unity implementation described in Section 3.4, so any advantage or defect in terrain rendering would have been shared by both the Unity application and its comparator rather than isolating the effect the study is designed to measure.

Kepler and the ArcGIS Online Experience Builder survive this elimination for complementary reasons. kepler.gl renders no terrain surface at all, which is not a shortcoming for the purposes of this study but a deliberate methodological control, since it allows the effect of terrain, argued for in Section 2.5 and implemented through the coordinate pipeline in Section 3.4, to be isolated as a variable rather than assumed. If the Unity application and the Experience Builder, both of

which render terrain, converge on a result that differs from kepler.gl, the difference can be attributed to terrain specifically rather than to some other property shared by two out of the three tools. The ArcGIS Online Experience Builder, in turn, represents the professional dashboard baseline discussed in Section 2.3, built with the same commercial infrastructure evaluated against Unity in Section 2.4, so that comparing against it tests whether the thesis platform offers anything beyond what a well resourced institutional GIS deployment already provides.

4.3.2 Scenario Selection and Comparability

The three scenarios were selected to be comparable in the dimensions that matter for the task set to participants, described in Section 4.5, rather than to be identical events. All three are reconstructed from MODIS active fire detections obtained through NASA FIRMS, the same sensor family and confidence model documented in Section 3.3.1, so that no cross tool difference in the results can be attributed to a difference in the underlying data model. All three are events of the same order of magnitude and set in mountainous terrain within protected areas, so that the terrain comprehension question in Section 4.1 is tested against comparably complex topography in every condition rather than against a flat landscape in one condition and a mountainous one in another.

As Table 2 shows, the three scenarios are not identical in scale. The kepler.gl scenario has the shortest window and the lowest detection count, at four days and 89 detections, while the Unity application's scenario is the longest and most detection-dense, at six days and 402 detections. The Experience Builder's scenario falls between the two on both measures, at six days and 349 detections. This asymmetry was accepted rather than eliminated because matching three real events exactly across all dimensions is not achievable without abandoning real-world data in favor of a synthetic scenario, which would have undermined the ecological validity of the comparison. It is treated explicitly as a limitation in Section 4.7 and is revisited when the comprehension results are interpreted in Chapter 6, since a longer window and a higher detection count could, in principle, make correct comprehension either easier, through more evidence, or harder, through more information to track, and the direction of this effect is not assumed in advance.

Table 2 The three evaluation scenarios and their properties within the window presented to participants

Tool	**Scenario**	**Window**	**Detections**	**Days with Detections**	**Burned area (ha)**
kepler.gl	Sierra de la Culebra, Spain	16–19 June 2022	89	Four	25,217
ArcGIS Online	Serra da Estrela, Portugal	6–11 August 2022	349	Six	≈ 25,000
Unity application	Jasper National Park, Canada	22 July – 23 August 2024	402	Six	32,722

Note: Data extracted from Copernicus Emergency Management Service (2022, 2023)

Pairing each tool with a distinct event is a deliberate methodological control in its own right, since presenting the same fire in all three conditions would allow participants to answer the comprehension questions in Section 4.4 from what they had learned in a previous condition rather than from what the tool being tested actually communicated, defeating the purpose of the comparison. This is also why the comprehension section of the questionnaire, described in Section 4.4, is scored only once, after the first tool each participant encounters, since it is only for that first tool that a participant's answers can be attributed to the tool's own communicative power rather than to accumulated familiarity with the general shape of the task.

It is noted that the clustering and marker scaling described in Section 3.6 are applied only within the Unity application, since kepler.gl and the Experience Builder render the extracted detections directly without any equivalent processing. This is not an oversight to be corrected but a necessary consequence of comparing three independently built tools rather than three configurations of the same tool, and it means that any advantage found for the Unity application on the intensity related items in Chapter 5 should be read as an advantage of that

specific encoding rather than as a claim that kepler.gl and the Experience Builder could not, in principle, implement something similar.

Finally, the three environments cannot be made strictly equivalent because they differ in camera behavior and interaction style, independent of the data or terrain they present. The study is therefore framed throughout as a comparison of user experience and communicative effectiveness rather than as a fully controlled experiment across equivalent environments. This framing is carried consistently into the data analysis plan in Section 4.6 and the limitations in Section 4.7.

4.4 The Questionnaire Instrument

Feedback is gathered with a structured questionnaire of 25 distinct questions in five parts: a background section, answered once, six comprehension questions, scored against a predefined answer key, ten usability items, six rating items on a five point scale, where one denotes strong disagreement and five denotes strong agreement, and three open questions, answered once at the end. This five part structure follows a deliberate ordering rather than an arbitrary one: the background section precedes any exposure to a tool so that self rated experience is not colored by how the study is going, the comprehension and usability and rating blocks are attached to each individual tool so that a participant's judgement of one tool cannot bleed into the recorded score for another, and the open questions are placed last, after every tool has been seen, so that free text reflection can draw on the full comparison rather than being anchored to whichever tool happened to be shown first.

The six comprehension questions map directly onto the first specific question set out in Section 4.1. Four are scored against a predefined answer key, one records the participant's perception of where fire activity concentrated and carries no correct answer, and one was excluded from scoring for the reason given in Section 4.7. The four scored items test the spatial location of the event, the day by day temporal progression, the number of detections displayed on a given day, as a check on whether the visualization supports quantitative reading rather than only qualitative impression, and the description of the surrounding terrain, which is the item most directly diagnostic of the terrain comparison described in Section 4.3.1.

Each scored item is keyed to what the tool in question presents to the participant. For the detection count item this differs from the archived total in the case of the Unity application,

which applies a stricter confidence threshold when data are loaded than when they are exported, as described in Section 3.5.

The usability section is based on the System Usability Scale, the same instrument used as an evaluation design precedent by Koutsaris (2023) and yields the familiar score ranging from zero to 100 derived from ten alternating positively and negatively worded statements. Four of its items were reworded during the design of the instrument to refer explicitly to a wildfire visualization rather than to a generic software system, and one of those four also changed polarity, so the scale no longer alternates strictly between positively and negatively worded statements. Scoring follows the wording of each item rather than its position in the published sequence. Because this rewording departs from the published wording of the scale, the value it yields is treated throughout the thesis as an adapted usability measure and is not compared with the published benchmark associated with the original scale, a distinction maintained consistently in Chapters 5 and 6.

The six rating items on the five-point scale address engagement and the four concrete visualization elements named in Section 4.1 (terrain context, the distinction between detections and effects, the interpretability of marker differences, and sustained attention), translating each research question in Section 4.1 into a single, directly answerable statement rather than leaving those questions to be inferred indirectly from the comprehension and usability scores alone. The three open questions, asked once at the end of the full sequence, invite free-text reflection on which tool best conveyed the event, which was easiest to use, and any remaining comments. They are used in Chapter 5 to provide qualitative context for the numerical patterns rather than as a separately scored measure.

The comprehension section is administered only once, after the first tool each participant encounters. As explained in Section 4.3.2, the underlying task structure (locate the area, follow the progression, identify the day of greatest activity) is learned after the first exposure, so a second or third comprehension score would reflect task familiarity as much as the tool itself. The usability and rating sections are administered after every tool, since a participant's judgment of how usable or engaging a specific interface is remains meaningful even after seeing other interfaces. This repeated measurement supports the background variable analysis described in Section 4.6.

4.5 Procedure and Online Delivery

The study is delivered entirely online to participants using six versions of a Google Form generated through Google Apps Script, one for each of the six possible presentation orders described below. All questions, the relevant answer key, and the tool-specific links are embedded directly in the corresponding form. Delivering the study this way removes any requirement for participants to install software or attend a session in person, consistent with the accessibility aim stated for the platform itself in Section 1.3, and allows recruitment to proceed through a single shared link per order rather than through scheduled sessions.

Each participant first reads a short description of the study and gives informed consent before any data are recorded, then provides the background information described in Section 4.2. The participant is then directed to the three tools one at a time, exploring each under a short guided task before completing that tool's questionnaire block and moving on to the next. The guided task asks the participant to locate the affected area, follow the fire across successive days, and identify the day of greatest activity, a fixed instruction given identically regardless of which tool is being viewed. This standardisation matters because, without a shared task, participants would be free to explore each tool in whatever way they found most natural, which would very likely differ between a terrain free tool such as kepler.gl and a fully navigable three dimensional environment such as the Unity application, confounding any resulting difference in comprehension or rating with a difference in how much attention participants happened to pay rather than with a difference in what the tool itself communicated.

The sequence in which the three tools are presented is counterbalanced across participants specifically to prevent the order effect that a fixed sequence would introduce, whereby whichever tool participants always saw first would receive an artificial comprehension advantage, since only the first tool is scored for comprehension, while whichever tool participants always saw last might receive an artificial usability or rating disadvantage from fatigue. Because three tools admit six possible orders, six versions of the questionnaire were prepared, one per order, and the corresponding links were distributed in rotation, in both English and Portuguese, which is later translated to English for normalization, to reach as wide a pool of nonspecialist participants as possible, so that approximately one third of participants began with each tool. No participant completed more than one version of the form, and participation was entirely self paced, with no fixed time limit imposed on any block.

4.6 Data Analysis Plan

Comprehension is compared between the three independent groups defined by the first tool encountered, since, as explained in Section 4.4, only the first tool provides a comprehension result free of the learning effect, this makes the comprehension comparison a between subjects design with three unrelated groups of 21, 24 and 23 participants respectively, rather than a repeated measures design. The usability measure and the six rating items are, for the same reason of comparability, also reported for the first tool encountered when the three tools are compared against each other directly, with differences between tools assessed using independent samples tests appropriate to three unrelated groups, reporting these two measures on the same first tool basis as comprehension, rather than pooling every participant's judgement of every tool, keeps the primary three way comparison methodologically consistent across all three constructs, comprehension, usability and rating, rather than mixing a between subjects comprehension result with a within subjects usability result.

The full repeated measures data, meaning every participant's usability and rating scores for all three tools rather than only the first one encountered, is retained separately and used to examine the influence of participant background. Because every participant rated every tool exactly once regardless of presentation order, pooling across viewing positions for this second analysis increases the number of observations available per background subgroup, at the cost of no longer distinguishing whether a score reflects a participant's first, second or third exposure. This trade off was accepted because the background analysis asks a different question from the main comparison, not which tool performs best overall, but whether the size of any usability gap depends on a participant's prior experience with geographic information systems or with three-dimensional environments, a question for which statistical power to detect a pattern across subgroups matters more than strict order control.

Given the modest sample described in Section 4.2, results throughout Chapter 5 and Chapter 6 are read as indicative evidence of comparative user experience rather than as statistically generalisable measurements. Differences between the three tools are nonetheless tested rather than asserted. Association between tool and accuracy on each scored comprehension item is assessed with a chi-square test of independence. Differences in the adapted usability score and in the rating items are assessed with the Kruskal-Wallis test, which does not assume normality and is appropriate to the ordinal scales and group sizes involved, followed by pairwise Mann-Whitney tests where the overall test is significant. No correction for the number of comparisons is applied, since the comparisons follow directly from the

questions set out in Section 4.1 rather than being selected after inspection of the data, and this is stated here so that the strength of language used in Chapter 5 can be judged against it.

4.7 Design Limitations and Ethical Considerations

Several limitations qualify how the results in Chapter 5 and their interpretation in Chapter 6 should be read and are stated here in advance rather than only in retrospect. The sample is a convenience sample rather than a probability sample of a defined population, as noted in Section 4.2, so findings indicate tendencies within the recruited group rather than values that can be generalised to a wider population without further study. The three tools are not strictly equivalent in interaction and camera behavior, as noted in Section 4.3.2, so part of any usability difference found in Chapter 5 may reflect general interface maturity and familiarity with a particular style of navigation rather than a property specific to wildfire visualization as such.

Each tool is paired with a distinct scenario rather than with the same event, a choice defended in Section 4.3.2 as necessary to avoid the learning effect, but this means a comprehension difference found between tools could, in principle, partly reflect a difference between the events themselves rather than between the tools presenting them, a risk mitigated but not eliminated by matching the three events on burned area and on mountainous protected area terrain as shown in Table 2. The kepler.gl environment renders no elevation surface at all, which, as explained in Section 4.3.1, is treated as a deliberate methodological control rather than as an incidental weakness, but it is nonetheless itself one of the properties under comparison rather than a neutral background condition, and this dual role should be kept in mind when the terrain related results are interpreted.

The usability instrument is an adapted measure, as detailed in Section 4.4, with four items reworded to fit a browser-based visualization rather than a generic recurring software system, so the resulting score is not comparable to published benchmarks associated with the unmodified scale. One comprehension item, concerning the direction of the fire's movement, offered response options that did not correspond meaningfully to the question as posed and was excluded from scoring after the fact rather than corrected retroactively, since amending a live instrument mid study would have introduced its own inconsistency across the six presentation orders, all other responses were retained exactly as submitted, including any that appear inconsistent with a participant's other answers.

5. Results

This chapter reports the outcomes of the user testing study described in Chapter 4 and the assessment of the application itself. In total, 68 participants completed the study: 21 began with kepler.gl, 24 with the ArcGIS Online Experience Builder, and 23 with the browser deployment of the Unity application. Comprehension is reported only for the first tool encountered, as explained in Section 4.4, and the usability and rating measures are likewise reported for that first tool, so that all three tools are compared on an equal footing. Statistical tests are reported where comparisons are made, following the analysis plan in Section 4.6. Given the sample size, non-parametric tests are used throughout for the ordinal measures, and the results are interpreted as indicative of comparative user experience rather than as generalizable population estimates.

5.1 Comprehension

The mean comprehension score out of four is 3.30 for the Unity application, 3.04 for the ArcGIS Online Experience Builder, and 2.43 for kepler.gl. The difference among the three groups is significant, as indicated by a Kruskal-Wallis test (H equals 9.41, two degrees of freedom, p equals 0.009).

Table 3 Correct responses to the scored comprehension items, by the first tool encountered, as a percentage of the group

Item	**kepler.gl**	**ArcGIS Online**	**Unity application**
Number of days	85.7	62.5	60.9
Detections on a stated date	61.9	58.3	82.6
Day of greatest activity	85.7	83.3	91.3
Terrain description	9.5	100.0	95.7

The terrain description item accounts for almost all of the difference and is the strongest single result in the study. Both terrain-based tools achieved near-perfect accuracy, 100 percent for the Experience Builder and 95.7 percent for the Unity application, compared with 9.5 percent for kepler.gl. The association between tool and accuracy on this item is very strong, with a chi-square of 54.68, two degrees of freedom, and a p-value below 0.001. The pattern of incorrect answers is as informative as their number: of the 19 kepler.gl participants who answered incorrectly, all selected flat lowland, an unambiguous description of a landscape that is in fact mountainous. No response option indicating uncertainty was selected by any participant in that group.

For the remaining three items, no significant association between tool and accuracy was found. The number of days item favored kepler.gl at 85.7 percent; chi-square equals 3.94, p equals 0.140, which is consistent with its being the shortest sequence at four days against six for the other two scenarios. The day of greatest activity was correctly identified by a large majority in every condition, with the Unity application highest at 91.3 percent, chi-square equals 0.68, p equals 0.712. The item asking how many detections were displayed on a stated date was answered correctly by 82.6 percent of Unity application participants, the highest of the three, against 61.9 percent and 58.3 percent; however, the association does not reach significance, chi-square equals 3.62, p equals 0.164, and no claim of a difference between tools is made on this item.

5.2 Usability and Ratings

Table 4 Adapted usability score and mean rating on the first tool encountered, with the standard error of each mean

Tool	n	Usability	Usability SE	Rating	Rating SE
kepler.gl	21	61.90	4.02	3.39	0.14

ArcGIS Online	24	74.38	3.40	4.08	0.12
Unity application	23	63.70	3.27	3.99	0.15

The three tools differ significantly on the adapted usability measure, with Kruskal-Wallis H equals 6.96, two degrees of freedom, p equals 0.031. Pairwise comparison locates the difference entirely with the Experience Builder, which scores higher than the Unity application, Mann-Whitney U equals 383.0, p equals 0.023, and higher than kepler.gl, U equals 155.5, p equals 0.029. The Unity application and kepler.gl do not differ, U equals 218.5, p equals 0.596. The usability advantage of the commercial platform is therefore a real result and is interpreted in Section 6.2.

The pattern on the rating measure is different. The three tools also differ, with H equals 13.92, p below 0.001. However, the Unity application and the Experience Builder are indistinguishable, U equals 295.0, p equals 0.692, while both are rated higher than kepler.gl, at p equals 0.002 and p below 0.001, respectively. The application is therefore judged as favorably as the commercial platform on the qualities the visualization was designed to deliver, while being judged less usable as an interface. Figure 14 places the three tools on both measures at once.

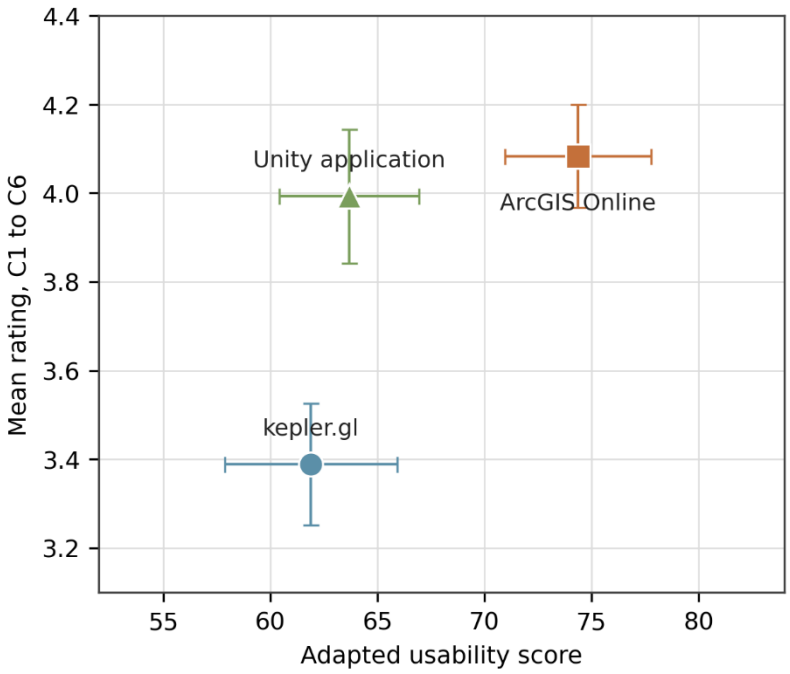


Figure 14 Adapted usability score plotted against mean rating for the three tools, first tool encountered, with standard error bars. Source: author.

Table 5 Mean rating per item on the first tool encountered, on a scale from one to five

Item	kepler.gl	ArcGIS Online	Unity application
Location could be identified	4.14	4.38	4.22
Day-to-day change could be followed	3.90	4.33	4.17
Terrain provided spatial context	2.52	4.17	4.30
Detections distinguished from effects	3.43	3.96	3.78

Item	kepler.gl	ArcGIS Online	Unity application
Meaning of marker differences was clear	3.10	3.83	3.30
Attention was held throughout	3.24	3.83	4.17

Two of the six items differ significantly across tools. Terrain context differs strongly, H equals 36.34, two degrees of freedom, p below 0.001, with the Unity application highest at 4.30 and the Experience Builder at 4.17, both far above kepler.gl at 2.52. Pairwise, each terrain-based tool exceeds kepler.gl at p below 0.001, while the two do not differ from each other, U equals 235.0, p equals 0.348. This converges with the objective terrain result in Table 3. Sustained attention also differs, H equals 10.45, p equals 0.005, with the Unity application highest at 4.17 against 3.83 and 3.24. Pairwise it exceeds kepler.gl, U equals 116.0, p equals 0.002, while the difference from the Experience Builder does not reach significance, U equals 206.0, p equals 0.119. Attention is therefore the item on which the application achieves its highest absolute rating and clearest separation from the terrain-free dashboard, rather than one on which it outperforms the commercial platform.

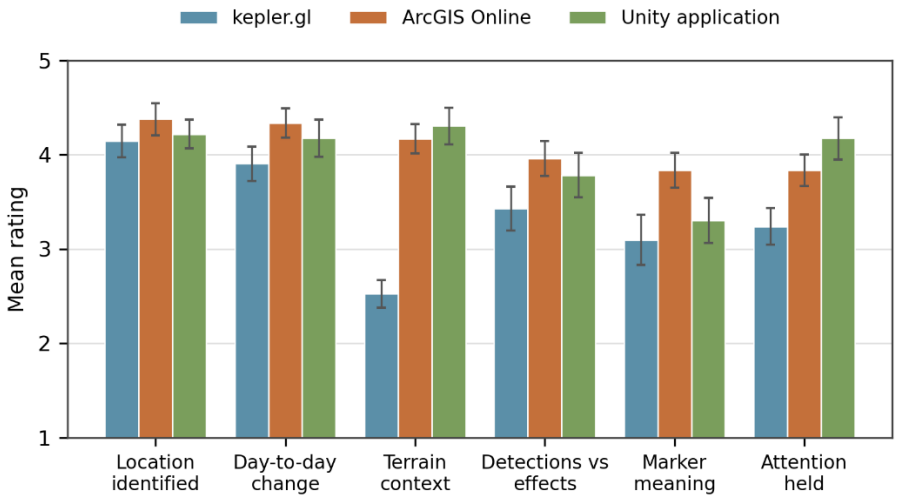


Figure 15 Mean rating of the three tools on each of the six visualization items, first tool encountered, with standard error bars. Source: author.

The remaining four items show no significant difference. Of these, the clarity of marker differences is the application's weakest item at 3.30, below the Experience Builder at 3.83, but the difference across the three tools does not reach significance, H equals 4.89, p equals 0.087. The item is therefore reported as the application's lowest-rated property rather than as a demonstrated deficit relative to the comparators, and it is returned to in Section 6.2. Figure 15 shows the profile of all three tools across the six items.

5.3 Influence of Participant Background

The usability and rating measures every participant provided for all three tools were related to the four background variables. Subgroups are small, three containing fewer than ten participants, so patterns are read as tendencies. The clearest pattern was that the Unity application’s usability rose steadily with participants that had prior experience, while its perceptual ratings remained stable across backgrounds.

Pooling across viewing positions is defensible here because the order check shows almost no position effect: mean adapted usability across the three positions is 66.91, 67.90 and 68.71, and mean rating is 3.84, 3.86 and 3.90, indicating that the counterbalancing described in Section 4.5 worked as intended.

Table 6 Adapted usability score by prior experience with geographic information systems, all viewing positions pooled, with group sizes

GIS experience	**n**	**kepler.gl**	**ArcGIS Online**	**Unity application**
None	27	56.9	77.4	53.1
Basic	10	64.0	74.8	66.5
Intermediate	15	55.3	78.0	65.5

GIS experience	n	kepler.gl	ArcGIS Online	Unity application
Advanced	16	72.8	86.2	75.0

Table 7 Adapted usability score by prior experience with three-dimensional or game environments, all viewing positions pooled, with group sizes

3D or game experience	n	kepler.gl	ArcGIS Online	Unity application
None or rare	23	62.4	80.4	56.8
Occasional	26	59.8	76.1	61.7
Frequent	19	62.1	82.1	72.1

Table 8 Adapted usability score by prior familiarity with wildfire or satellite fire detection data, all viewing positions pooled, with group sizes

Familiarity	n	kepler.gl	ArcGIS Online	Unity application
None	39	57.1	74.9	56.9
Some	22	65.0	84.1	70.3
Considerable	7	73.2	88.2	73.9

The clearest trend concerns the Unity application. Its usability score rises across every level of both experience variables, from 53.1 among participants reporting no experience with geographic information systems to 75.0 among those reporting advanced experience, and from

56.8 among those with no or rare experience with three-dimensional environments to 72.1 among frequent users. The Experience Builder shows no comparable gradient, scoring between 74.8 and 86.2 across all levels of both variables, and kepler.gl shows no consistent gradient at all. Figure 16 shows the three-dimensional experience pattern.

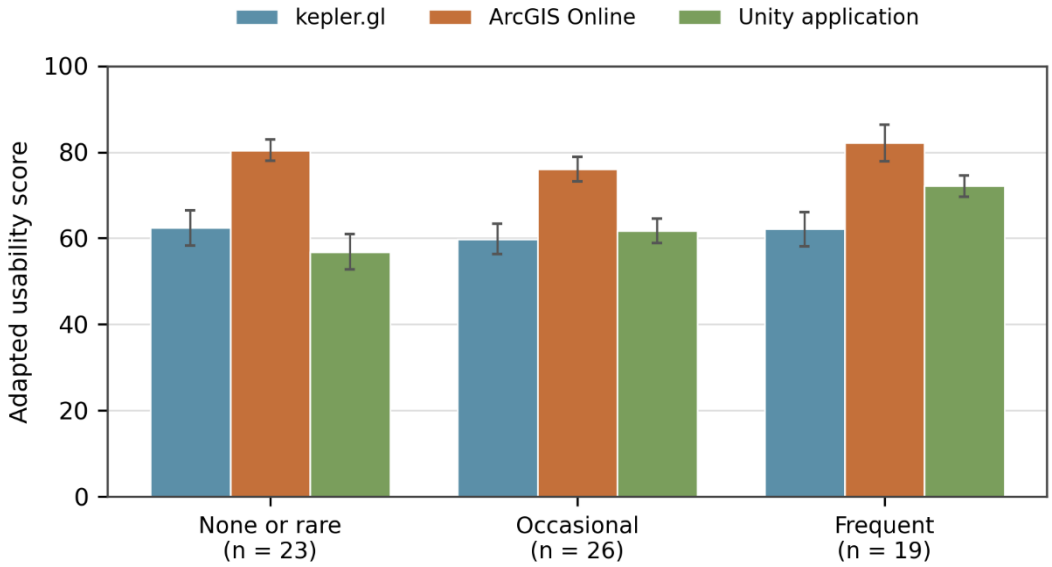


Figure 16 Adapted usability score by prior experience with three-dimensional or game environments, all viewing positions pooled, with group sizes and standard error bars. Source: author.

Table 9 Adapted usability score by field of study, all viewing positions pooled, with group sizes

Field of study	**n**	**kepler.gl**	**ArcGIS Online**	**Unity application**
Environmental or earth sciences	15	60.2	82.2	65.8
Geoinformatics or GIS	17	72.9	84.9	71.8
Computer science or software	9	53.9	78.1	74.4

Field of study	n	kepler.gl	ArcGIS Online	Unity application
Communication, media, or design	9	57.2	75.3	53.6
Other	18	57.1	74.0	51.2

The gap between the application and the commercial platform is therefore not constant across the sample. It is widest among participants with no relevant prior experience, at 24.3 points on the GIS variable and 23.6 points on the three-dimensional variable and narrows to 11.2 and 10.0 points respectively among the most experienced. Crucially, the rating measures show no equivalent gradient, which indicates that participants who found the application harder to operate were nonetheless able to read what it displayed. This distinction is central to the interpretation offered in Section 6.2.

5.4 Open-Ended Responses

After viewing all three tools, participants answered three open questions: which aspects of the visualizations they liked most, which were most confusing, and what changes would most improve the experience. All 68 participants responded, in English or Portuguese, with Portuguese responses translated for analysis. What follows groups recurring themes rather than reproducing every response, and where a participant referred to a tool by position rather than by name, the reference was resolved against that participant's presentation order before coding.

Two positive themes dominate irrespective of which tool a participant encountered first. The day-by-day time lapse was named by 31 of the 68 participants, making it the single most frequently praised feature across all three tools, and terrain representation was named by 27. Both directly support the design propositions set out in Section 2.6 and Section 3.6, and the terrain responses mirror the objective terrain result in Table 3, with five participants explicitly identifying the absence of an elevation surface in kepler.gl as a shortcoming.

The most frequent criticism concerns navigation in the Unity application. Eighteen participants described the camera or movement scheme as difficult, a volume of complaint with no equivalent for either comparator. Their responses fall into three distinct complaints. Several described the scheme as hard to learn. Several specifically wanted the conventional mouse-drag-and-scroll interaction of a map application rather than the keyboard-driven scheme offered, with one noting that only the WASD keys functioned for movement. A third group described losing orientation or having to move a considerable distance before locating the fire, prompting several requests that the camera open on the fire or follow it automatically. This pattern is consistent with and helps explain the experience gradient in Section 5.3: participants unfamiliar with three-dimensional navigation spent part of the session learning to move through the scene, which the usability instrument records and the rating instrument does not.

Nine participants reported technical problems. Seven of these concerned the Unity application, including longer loading times than the other two tools, an error message that was resolved by refreshing, a need to refresh repeatedly, one crash on first use, and one report that the application did not run acceptably on the participant's laptop. This is a distinct category from the interaction difficulties above and is examined further in Section 5.5. One participant reported a comparable failure with kepler.gl, which did not run in their browser, and another reported a problem across all three tools.

Eight participants asked for a legend or reported that the meaning of marker size or symbology was unclear, most referring to the Unity application. Two specifically stated they could not tell what the differences in point size meant, and one asked that the wind arrows be labeled. This corroborates, on qualitative grounds, the lowest-rated item in Table 5 and forms the basis of the recommendation in Section 6.2. Notably, a temperature legend for the wind arrows is already present in both builds, so the requirement identified here is specifically for a legend explaining fire radiative power and marker scaling.

Four participants commented on the daily detection counter, either praising its immediacy in tools that provide one or requesting that kepler.gl provide an equivalent. This is relevant to the interpretation of the detection count item in Section 5.1, as it indicates that the counter serves as a usable enumeration channel that participants notice. Kepler.gl drew recurring criticisms beyond the absence of terrain, chiefly that its interface was too dark and low in contrast against the fire symbols, and that its time control was difficult to read and operate precisely. The Experience Builder drew comparatively few complaints, the most

frequent being that its fire symbols were too small, which is consistent with its highest usability score in Table 4. Four participants reported no difficulties, proposed no changes, or didn't know what could be changed in any of the three tools.

5.5 Comparison of the Desktop and Browser Builds

The application described in Chapter 3 is the deliverable of this thesis, and the browser deployment is the instrument through which it was made reachable for the comparison in Sections 5.1 to 5.4. This section reports what the application renders, how the two builds differ (Table 10).

Table 10 Capability of the desktop reference build and the browser deployment for the Jasper scenario

Property	**Desktop build**	**Browser build**
Distribution	Self contained Windows executable	Static host with cross-origin isolation headers
Minimum requirement	Windows with dedicated graphics hardware	Browser supporting WebGL 2.0 and SharedArrayBuffer
Fire data format	Shapefile via NetTopologySuite	JSON via a custom parser
Launch configuration	Fire, weather and cloud inputs, start location, render preset	None exposed
Save and load visualization state	Present	Absent
Render preset switching	Quality and Performance	Performance

Property	Desktop build	Browser build
Timeline span	33 days, detections on six	33 days, detections on six
Confidence threshold at load	High confidence band	High confidence band
Fire markers, smoke, burn scars	Present	Only markers
Wind arrow grid with temperature legend	Present	Present
Cloud cover and detection counter	Present	Present
Terrain conforming	Raycast snap with retries	As desktop, plus a recurring re-snap pass
Interactive minimap with compass	Present	Absent
Cartographic scale bar	Present	Present

The desktop build includes the full feature set, and its launch configuration allows a new study area to be visualized without recompiling, which is the basis for the reproducibility claim in Section 1.3. The browser build removes that configuration, the minimap and compass, the preset switching, and the smoke and burn scar layers. It is not a strict subset, however, because progressive terrain streaming requires a recurring pass that re-snaps drifted markers,

which the desktop build does not need. In both builds, the timeline spans the full acquisition range requested from FIRMS, 33 days for the Jasper scenario, of which six carry active detections, and the remainder are empty. The slider therefore steps through the entire period rather than only the days when the fire was detected.

Seven of the 68 participants reported application-specific problems: longer loading times than the comparators, an error message that was resolved by refreshing, repeated refreshes, one crash, and one instance where it failed to run acceptably on a laptop. The browser build requires cross-origin isolation via two HTTP response headers; without them, the globe renderer cannot initialize, and it must download its data into browser storage before any reading begins.

Eighteen participants described the camera or movement scheme as difficult, with no comparable volume for either comparator. This is an interaction design fault rather than a technical one, since the reported rating measures in Section 5.3 remain stable across all experience levels.

A less severe issue appears in the timeline control, where the comprehension item received the lowest score at 60.9 percent. This is probably because the slider covers 33 days, with detections only on six of those days, leading the control to produce two reasonable interpretations of the same question.

The tool participants evaluated presented a reduced subset of the recorded activity, excluding the smoke and burn scar layers and omitting any configuration or navigation aids from the desktop build. The results for the application are therefore a lower bound on what the pipeline communicates rather than a measurement of the deliverable. The desktop build was not tested with users because requiring installation would have broken comparability with two browser-based comparators, and that comparison is the most immediate extension of this work.

6. Discussion

This chapter interprets the results reported in Chapter 5 against the objectives set out in Section 1.3, situates them in relation to the work reviewed in Chapter 2, and states the limitations that qualify them.

6.1 Objectives Revisited

The first objective, a workflow visualizing wildfire extent and temporal progression from MODIS detections, is met for progression and qualified for extent in both applications. Temporal progression is supported directly. The day of greatest activity was identified by 91.3 percent of participants using the application, the highest of the three tools, and the ability to follow day-to-day change was rated 4.17 (Table 5). The animated time lapse was the most frequently praised feature in the open responses, cited by 31 of the 68 participants.

Extent requires qualification. As documented in Section 3.5, a stricter confidence threshold is applied when data are loaded than when they are exported, so the scene presents fewer detections than the archive records, with the loss concentrated at the margins of the front where radiative power and confidence are both lowest. The application therefore accurately represents the position and shape of the event and conservatively estimates its full extent.

The second objective, enabling users to identify spatial and temporal patterns and navigate an immersive three-dimensional environment, is met for identification and engagement but not for navigation by inexperienced users.

Spatial identification is strong: 95.7 percent described the terrain correctly and the terrain context item was rated 4.30, the highest of the three tools, with the difference across tools significant. Engagement is the item on which the application achieves its highest absolute rating, at 4.17, and the three tools differ significantly on it. Navigation is the exception, and it is a clear one. Eighteen participants described the camera or movement scheme as difficult, with no comparable volume for either comparator, and the adapted usability score rises steadily with prior experience of three-dimensional environments, from 56.8 among participants with none or rare experience to 72.1 among frequent users (Table 7). The objective is therefore met for what the environment communicates and not met for how easily a novice moves through it.

The third objective, a pipeline implemented with freely licensed components and reproducible without a programming background, is not addressed by the user study, which tested end users viewing outputs rather than practitioners rebuilding the pipeline. It remains supported by the implementation in Chapter 3 rather than by testing.

The claim requires one qualification: the pipeline requires no paid license and avoids proprietary GIS platforms, which is the substantive claim and holds. It is not, however, composed exclusively of open-source components: the fire and smoke effects are third-party assets distributed under licenses that do not permit redistribution, and the terrain and imagery are streamed from a hosted service whose free tier excludes commercial use. A practitioner can therefore rebuild the pipeline at no cost but cannot obtain a working build from the source repository alone without separately acquiring those assets. This is stated here rather than left for a reader to discover.

The fourth objective, establishing Unity3D as a viable geovisualization platform, is supported by the evidence in Chapter 5. The application produced the highest mean comprehension score of the three tools, 3.30 against 3.04 and 2.43, and was rated as favorably as the commercial platform on the qualities the visualization was designed to deliver, with no significant difference between them on the rating measure. A game engine, used with a georeferenced terrain pipeline, therefore matched a mature commercial GIS deployment in communicative effectiveness in this comparison, which is the claim the objective requires.

6.2 Principal Findings

Terrain is a precondition for correct spatial understanding rather than a presentational enhancement. This is the strongest result in the study and the clearest support for the thesis's central proposition. Participants using the tool without a rendered elevation surface described a mountainous protected area correctly in 9.5 percent of cases, against 100 percent and 95.7 percent for the two terrain-based tools, an association of exceptional strength (chi-square equals 54.68, p below 0.001).

Communicative effectiveness and interface usability separate cleanly, and the application succeeds on the first while trailing on the second. The application and the commercial platform are statistically indistinguishable on the rating measure, p equals 0.692, and on all six rating items taken individually, where the smallest pairwise p value is 0.119,

while the commercial platform scores significantly higher on the adapted usability measure, p equals 0.023. The application is therefore not a weaker communicator that participants tolerated; it is an equally effective communicator that participants found harder to operate. Reporting a single satisfaction score would have concealed this, which vindicates the decision in Section 4.1 to keep the three constructs separate.

The usability gap is an onboarding problem, it is localized in navigation, and it is not uniform across the sample. Table 6 and Table 7 show the application's usability rising across every level of both experience variables while the commercial platform stays flat, so the gap is widest for participants with no relevant background, at 23.6 points on the three-dimensional variable, and narrows to 10.0 points among frequent users. Critically, the rating measures show no equivalent gradient. Participants who struggled to move through the scene were still able to read what it showed. The open responses locate the difficulty precisely: the keyboard-driven camera, the absence of the mouse drag and scroll interaction participants expected from a map application, and the effort required to locate the fire on entry. All three are addressable through interface design without altering the rendering pipeline, and the "Go to Fire" control that already exists suggests the remedy is a default behavior rather than a new feature.

Sustained attention is the application's distinctive advantage. The application records its highest absolute rating on this item, at 4.17, and the three tools differ significantly, p equals 0.005, with the application clearly separated from the terrain-free dashboard although not from the commercial platform. Taken with the frequency of unprompted praise for the time lapse, this supports the narrative cartography argument of Section 2.6 empirically rather than by assertion: the combination of animated progression, particle effects and terrain context holds a non-specialist audience in a way that a conventional dashboard does not.

The visualization under-reports what it displays, and the viewer is not told. The clarity of marker differences was also the application's lowest-rated property, at 3.30 (Table 5), and eight participants asked without prompting for a legend explaining it. The application applies a stricter confidence threshold at load than the archive carries and then merges proximate detections into single scaled markers, so three different quantities coexist in one scene: what was recorded, what was loaded, and what is visible. Only the second is reported, in the timeline overlay, and only four participants referred to that counter in the open responses. Two remedies follow, both inexpensive: report the recorded count alongside the displayed count in the overlay, and provide an explicit fire radiative power legend for the markers. A temperature

legend for the wind arrows is already present in both builds, so the requirement identified by participants is specifically for the fire marker encoding rather than for legends in general.

6.3 Related Work and Limitations

The terrain result extends rather than confirms the existing literature. Evangelidis et al. (2018) and Laksono and Aditya (2019) establish that three-dimensional geospatial visualization and game engine environments improve the immersive quality and real-world fidelity of spatial data presentation, and Thöny et al. (2018) identify contextual framing, the embedding of dynamic event layers within static terrain, as one of three mechanisms of effective three-dimensional geographic storytelling. The present study measures the consequence of removing that mechanism and finds it to be not a reduction in engagement but a systematic and confident factual error about the landscape itself. That is a stronger and more specific claim than the general assertion that terrain aids understanding.

The findings also sharpen the distinction drawn in Section 2.3 between simulation-first and visualization-first tools. Wu et al. (2023) demonstrate that GPU-accelerated fire-spread simulation can achieve substantial computational gains while remaining oriented toward technical users, and Tirado Cortes et al. (2025) extend immersive interfaces for trained responders. The present work addresses the complementary case, an observational reconstruction aimed at audiences with no domain training, and the results indicate that the communicative gap identified by Inglis and Vukomanovic (2020) between data-rich dashboards and public understanding is addressable with the visual apparatus of a game engine. Buyuksalih et al. (2017) noted limitations in handling large shapefile datasets natively within Unity.

The evaluation design follows the combination of a usability scale with visualization-specific rating items established by Koutsaris (2023), extended here in three ways: an objective, scenario-specific comprehension component scored against an answer key, a three-tool comparison rather than a single-system assessment, and a counterbalanced between-subjects allocation that isolates the first exposure from learning effects.

Six limitations qualify the findings above. The sample of 68 is a convenience sample rather than a probability sample, and three background subgroups contain fewer than ten

participants, so the patterns in Section 5.3 are read as tendencies rather than as tested effects and no inferential test is reported for them.

Each tool is paired with a distinct wildfire event, a choice defended in Section 4.3.2 as necessary to defeat the learning effect, which means a comprehension difference could in principle reflect the event rather than the tool. This is mitigated but not eliminated by matching the three events on order of magnitude of burned area and on mountainous protected area terrain.

The three environments differ in camera behavior and interaction style independently of the data they present, so part of the usability difference may reflect general interface maturity rather than a property specific to wildfire visualization.

The build evaluated was the browser deployment rather than the desktop reference build, for the reason given in Section 3.9, and it presents a reduced subset of the recorded activity. The results are therefore best read as a lower bound on what the pipeline communicates rather than as a measurement of the deliverable itself.

One comprehension item was excluded before scoring because its response options did not correspond to the question as posed, as disclosed in Section 4.7. The usability instrument is an adapted measure with four items reworded, one of which also changed polarity, so the resulting score is not comparable to published benchmarks associated with the unmodified scale.

Finally, two components of the implementation were not evaluated at all. The meteorological integration described in Sections 3.3.2 and 3.7, comprising wind-driven marker animation and the wind arrow grid, forms a substantial part of the methodology but was not assessed by any comprehension or rating item, and no measured rendering benchmark comparing the two builds was conducted. Both are identified as future work in Chapter 7 rather than presented as results.

7. Conclusion

The aim of this thesis was to design and implement a modular Unity3D application capable of transforming satellite fire detection records into an immersive, geographically accurate three-dimensional time lapse without dependence on any proprietary paid solution,

and to test whether such a tool communicates a wildfire event to a non-specialist audience more effectively than the two-dimensional dashboards and commercial GIS platforms currently available. The application was delivered in two builds: a desktop reference implementation carrying the framework at full capability, and a browser deployment of the same pipeline that requires no installation and through which the comparative evaluation was conducted.

The principal finding of the evaluation concerns terrain, and it is the clearest result the study produced. Participants shown the same class of event in a tool that renders no elevation surface described a mountainous protected area correctly in 9.5 percent of cases, against 100 percent and 95.7 percent for the two terrain-based tools. The significance of this lies less in the size of the difference than in the character of the errors: every participant who answered incorrectly selected flat lowland, and none selected a response expressing uncertainty. The absence of terrain did not leave viewers hesitant about the landscape, it left them confidently and uniformly mistaken about it. Terrain is therefore established here not as a presentational enhancement but as a precondition for correct spatial understanding, and the substantial engineering commitment the pipeline makes to streaming and conforming a global elevation surface is vindicated by that result rather than by argument.

On the measures the application was designed to serve, it performed well. It produced the highest mean comprehension score of the three tools, 3.30 out of four against 3.04 for the commercial platform and 2.43 for the terrain-free dashboard. It led on identifying the day of greatest fire activity and on holding participants' attention, the latter being the item on which it recorded its highest absolute rating. On the rating measure overall, it was statistically indistinguishable from the commercial platform, so it was judged as effective a communicator of the event as a mature institutional GIS deployment.

Where it did not perform well is equally specific and is stated here rather than absorbed into a general claim of success. The application was found significantly less usable than the commercial platform, and that gap is concentrated among participants with no prior experience of geographic information systems or of three-dimensional environments, narrowing by more than half among experienced users. The open responses locate the difficulty in navigation rather than in the visualization: the keyboard-driven camera, the absence of the mouse interaction participants expected from a map application, and the effort of locating the fire on entry. Notably, the rating measures show no equivalent gradient, which indicates that

participants who struggled to move through the scene were nonetheless able to read what it displayed. The gap is an onboarding problem, not a communication problem.

Two further weaknesses were identified. The meaning of the fire marker encoding was the application's lowest-rated property, and participants asked without prompting for a legend explaining it. More seriously, the application presents fewer detections than the archive records, because a stricter confidence threshold is applied when data are loaded than when they are exported, and it then merges proximate detections into single scaled markers. Three quantities therefore coexist in one scene, what was recorded, what was loaded and what is visible, and only the second is reported to the viewer.

The thesis contributes a reproducible pipeline that converts satellite-observed fire data into a three-dimensional narrative intelligible to educators, journalists and the public; an open bridge between .NET geoinformatics libraries and the Unity runtime through NuGet, allowing shapefiles to be read and georeferenced at runtime; and empirical evidence, from a counterbalanced three-tool comparison with 68 participants, that a game engine can match a commercial GIS platform on communicative effectiveness while remaining freely available. The pipeline requires no paid license and avoids proprietary GIS platforms, although it is not composed exclusively of open-source parts, since the fire and smoke effects and the terrain streaming service carry licenses that restrict redistribution and commercial use.

Four directions follow directly from the results. The navigation scheme should be reworked to match the conventional map interaction participants expected, with the camera framing the fire on entry by default, which the evidence suggests would close most of the usability gap without altering the visualization. An explicit fire radiative power legend should be added, alongside a display of the recorded detection count next to the displayed count, so that what the scene omits is visible to the viewer. The meteorological component, comprising wind-driven marker animation and the wind arrow grid, is fully implemented but was not assessed by any item in the evaluation, and deserves a study of its own to determine whether animated environmental context aids or distracts from comprehension of the fire itself. Finally, the desktop build, which carries the complete feature set and the unreduced dataset, has not been evaluated with users, and comparing it against the browser deployment would establish how much of the result reported here is attributable to the pipeline and how much to the constraints of a browser runtime.

What this thesis demonstrates is narrower than a general claim of success and more useful than one. A freely available game engine pipeline, given real terrain and an animated daily progression, communicates where a wildfire happened and how it developed at least as well as a commercial platform and considerably better than a flat dashboard, and it holds attention better than either. It does so while being harder to operate for viewers without relevant experience, and while showing fewer detections than it records. Both of those are specific, measured and addressable, which is a stronger position from which to continue the work than an unqualified claim would have been.

References

Figures

[1] United States Environmental Protection Agency. (2022). Climate change indicators: Wildfires. Retrieved from: https://www.epa.gov/climate-indicators/climate-change-indicators-wildfires

[2] Copernicus Atmosphere Monitoring Service. (2024, December 4). CAMS global wildfires review 2024: A harsh year for the Americas. Retrieved from: https://atmosphere.copernicus.eu/cams-global-wildfires-review-2024-harsh-year-americas

Literature

BARLOW, J., BERENGUER, E., CARMENTA, R., FRANÇA, F. (2020): Clarifying Amazonia's burning crisis. *Global Change Biology*, 26(2), pp. 319–321. doi:10.1111/gcb.14872

BUYUKSALIH, I., BAYBURT, S., BUYUKSALIH, G., BASKARACA, A. P., KARIM, H., RAHMAN, A. A. (2017): 3D modeling and visualization based on the Unity game engine: advantages and challenges. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, IV-4/W4, pp. 161–166.

CAQUARD, S., CARTWRIGHT, W. (2014): Narrative cartography: From mapping stories to the narrative of maps and mapping. *The Cartographic Journal*, 51(2), pp. 101–106.

CARTWRIGHT, W., FIELD, K. (2015): Exploring cartographic storytelling: Reflections on mapping real-life and fictional stories. In: *Proceedings of the 27th International Cartographic Conference (ICC 2015)*. Rio de Janeiro, Brazil.

CESIUM (2025): Cesium for Unity [online]. Philadelphia: Cesium GS [cit. 2026-07-24]. Available online: https://cesium.com/platform/cesium-for-unity/.

COPERNICUS ATMOSPHERE MONITORING SERVICE (2024): CAMS global wildfires review 2024: A harsh year for the Americas [online]. Reading: ECMWF [cit. 2026-07-24]. Available online: https://atmosphere.copernicus.eu/cams-global-wildfires-review-2024-harsh-year-americas.

NASA FIRMS (2026): Fire Information for Resource Management System [online]. Greenbelt: NASA [cit. 2026-07-24]. Available online: https://firms.modaps.eosdis.nasa.gov/.

UNITED STATES ENVIRONMENTAL PROTECTION AGENCY (2022): Climate change indicators: Wildfires [online]. Washington: EPA [cit. 2026-07-24]. Available online: https://www.epa.gov/climate-indicators/climate-change-indicators-wildfires.

CUNNINGHAM, C. X., WILLIAMSON, G. J., BOWMAN, D. M. J. S. (2024): Increasing frequency and intensity of the most extreme wildfires on Earth. *Nature Ecology & Evolution*, 8, pp. 1420–1425. doi:10.1038/s41559-024-02452-2

EVANGELIDIS, K., PAPADOPOULOS, T., PAPATHEODOROU, K., MASTOROKOSTAS, P., HILAS, C. (2018): 3D geospatial visualizations: Animation and motion effects on spatial objects. *Computers & Geosciences*, 111, pp. 200–212. doi:10.1016/j.cageo.2017.11.007

GIGLIO, L., SCHROEDER, W., JUSTICE, C. O. (2016): The Collection 6 MODIS active fire detection algorithm and fire products. *Remote Sensing of Environment*, 178, pp. 31–41. doi:10.1016/j.rse.2016.02.054

GUO, Z., LI, W., CIAIS, P., SITCH, S., VAN DER WERF, G. R., BOWRING, S. P. K., BASTOS, A., MOUILLOT, F., HE, J., SUN, M., ZHU, L., DU, X., WANG, N., HUANG, X. (2025): Reconstructed global monthly burned area maps from 1901 to 2020. *Earth System Science Data*, 17, pp. 3599–3618.

HANSEN, R. (2021): Bridging the world of 3D GIS and game engines. *ArcUser* [online]. Redlands: Esri [cit. 2026-07-24]. Available online: https://www.esri.com/about/newsroom/arcuser/bridging-the-world-of-3d-gis-and-game-engines.

HUANG, T., FENG, Z., PAES, D., YING, F., ZHAO, X., KINATEDER, M., LANGER, E. R., LOVREGLIO, R. (2025): Gamification for wildfire education and safety training: A systematic literature review and meta-analysis. *Journal of Computer Assisted Learning*, 41(5), e70108.

HUANG, Y., LI, J., ZHENG, H. (2024): Modeling of wildfire digital twin: Research progress in detection, simulation, and prediction techniques. *Fire*, 7(11), 412.

INGLIS, N. C., VUKOMANOVIC, J. (2020): Visualizing when, where, and how fires happen in U.S. parks and protected areas. *ISPRS International Journal of Geo-Information*, 9(5), 333. doi:10.3390/ijgi9050333

KEELEY, J. E., PAUSAS, J. G. (2022): Evolutionary ecology of fire. *Annual Review of Ecology, Evolution, and Systematics*, 53, pp. 203–225. doi:10.1146/annurev-ecolsys-102320-095612

KOUTSARIS, A. (2023): *Immersive visualization of real forests before and after wildfires*. Master's thesis. Utrecht: Utrecht University. Available online: https://studenttheses.uu.nl/handle/20.500.12932/44889.

LAKSONO, D., ADITYA, T. (2019): Utilizing a game engine for interactive 3D topographic data visualization. *ISPRS International Journal of Geo-Information*, 8(8), 361. doi:10.3390/ijgi8080361

MENG, Q., LU, H., HUAI, Y., XU, H., YANG, S. (2023): Forest fire spread simulation and fire extinguishing visualization research. *Forests*, 14(7), 1371.

NASA FIRMS (2026): Fire Information for Resource Management System [online]. Greenbelt: NASA [cit. 2026-07-24]. Available online: https://firms.modaps.eosdis.nasa.gov/.

PARKS CANADA (2024): Wildfire status: Jasper National Park [online]. Jasper: Parks Canada [cit. 2026-07-24]. Available online: https://parks.canada.ca/pn-np/ab/jasper/visit/feu-alert-fire/feudeforet-wildfire.

PAVELKA, K., Jr., LANDA, M. (2024): Using virtual and augmented reality with GIS data. *ISPRS International Journal of Geo-Information*, 13(7), 241. doi:10.3390/ijgi13070241

SUN, Z., LO, J., HU, J. (2026): WildfireX-SLAM: A large-scale low-altitude RGB-D dataset for wildfire SLAM and beyond. In: LOKOČ, J. et al. *MultiMedia Modeling (MMM 2026)*, Lecture Notes in Computer Science, Vol. 16413. Singapore: Springer. doi:10.1007/978-981-95-6957-1_38

THÖNY, M., SCHNÜRER, R., SIEBER, R., HURNI, L., PAJAROLA, R. (2018): Storytelling in interactive 3D geographic visualization systems. *ISPRS International Journal of Geo-Information*, 7(3), 108. doi:10.3390/ijgi7030108

TIME (2025): Inside the mega-fires: How wildfires are creating their own weather [online]. New York: Time [cit. 2026-07-24]. Available online: https://time.com/7299284/age-of-mega-wildfires-climate-change/.

TIRADO CORTES, C. A., THUROW, S., ONG, A., SHARPLES, J. J., BEDNARZ, T., STEVENS, G., DEL FAVERO, D. (2024): Analysis of wildfire visualization systems for research and training: Are they up for the challenge of the current state of wildfires? *IEEE Transactions on Visualization and Computer Graphics*, 30(7), pp. 4285–4303. doi:10.1109/TVCG.2023.3258440

TIRADO CORTES, C. A., FLORES GONZALEZ, M., HURLEY, R. J., DEL FAVERO, D. (2025): Designing an immersive wildfire visualization system for firefighters. In: *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, pp. 1–10.

TRENHOLME, D., SMITH, S. P. (2008): Computer game engines for developing first-person virtual environments. *Virtual Reality*, 12(3), pp. 181–187.

UNITED STATES ENVIRONMENTAL PROTECTION AGENCY (2022): Climate change indicators: Wildfires [online]. Washington: EPA [cit. 2026-07-24]. Available online: https://www.epa.gov/climate-indicators/climate-change-indicators-wildfires.

UNITY TECHNOLOGIES (2025): Timeline package documentation [online]. San Francisco: Unity Technologies [cit. 2026-07-24]. Available online: https://docs.unity3d.com/Packages/com.unity.timeline@latest.

WU, R., SCULLY-ALLISON, C., CARTHEN, C., GARCIA, A., HOANG, R., LEWIS, C., QUIJADA, R. S., SMITH, J., DASCALU, S. M., HARRIS, F. C., Jr. (2023): vFirelib: A GPU-based fire simulation and visualization tool. *SoftwareX*, 23, 101411. doi:10.1016/j.softx.2023.101411

YANG, X., URBAN, M. C., SU, B. et al. (2026): Wildfire risk for species under climate change. *Nature Climate Change*, 16, pp. 613–621. doi:10.1038/s41558-026-02600-5

List of Attachmets

Attachment 1 Source code repository:
https://github.com/marcelorigoto1/wildfire-visualization

Attachment 2 Desktop application, Windows build:
https://drive.google.com/file/d/1qgj2AZptZzWIgo7oWqSjmxBNU8DlRDjz/view

Attachment 3 Browser deployment of the application:
https://beamish-gelato-46180a.netlify.app

Attachment 4 Study response data:
https://docs.google.com/spreadsheets/d/1hdP6mA_PqQosdb_e_qs4wc2l47dAdIfS/edit?usp=sharing

All four attachments are provided in digital form. Attachments 1, 2 and 4 are also submitted to the department on a storage medium together with the thesis text.

Appendix

Appendix A: Supporting Study Data

This appendix reports the consolidated data underlying Chapter 5. It is provided so that the results reported there can be checked without recourse to the response file, which is available in full as Attachment 4. Per-participant responses are not reproduced here, since 68 participants across 25 questions is not legible in print.

Table A1 Composition of the sample, 68 participants

Variable	**Category**	**n**	**%**
Age group	18 to 24	19	27.9
	25 to 34	34	50
	35 to 44	12	17.6
	45 to 54	3	4.4
Field of study or background	Environmental or earth sciences	15	22.1
	Geoinformatics or GIS	17	25
	Computer science or software	9	13.2
	Communication, media, or design	9	13.2
	Other	18	26.5

Variable	Category	n	%
Experience with geographic information systems	None	27	39.7
	Basic	10	14.7
	Intermediate	15	22.1
	Advanced	16	23.5
Experience with three-dimensional or game environments	None or rare	23	33.8
	Occasional	26	38.2
	Frequent	19	27.9
Familiarity with wildfire or satellite fire detection data	None	39	57.4
	Some	22	32.4
	Considerable	7	10.3

Table A2 Complete results of the tests reported in Chapter 5, first tool encountered

Test	Measure	Statistic	df	p
Chi-square	Number of days	3.94	2	0.140

Test	Measure	Statistic	df	p
Chi-square	Detections on a stated date	3.62	2	0.164
Chi-square	Day of greatest activity	0.68	2	0.712
Chi-square	Terrain description	54.68	2	below 0.001
Kruskal-Wallis	Comprehension score, of four	9.41	2	0.009
Kruskal-Wallis	Adapted usability score	6.96	2	0.031
Kruskal-Wallis	Rating mean, C1 to C6	13.92	2	below 0.001
Kruskal-Wallis	C1 Location could be identified	1.43	2	0.489
Kruskal-Wallis	C2 Day-to-day change could be followed	3.21	2	0.201
Kruskal-Wallis	C3 Terrain provided spatial context	36.34	2	below 0.001
Kruskal-Wallis	C4 Detections distinguished from effects	3.65	2	0.161
Kruskal-Wallis	C5 Meaning of marker differences was clear	4.89	2	0.087
Kruskal-Wallis	C6 Attention was held throughout	10.45	2	0.005

Table A3 Pairwise Mann-Whitney comparisons, first tool encountered

Measure	Comparison	U	p
Adapted usability score	kepler.gl vs ArcGIS Online	155.5	0.029
	kepler.gl vs Unity application	218.5	0.596
	ArcGIS Online vs Unity application	383.0	0.023
Rating mean, C1 to C6	kepler.gl vs ArcGIS Online	102.5	below 0.001
	kepler.gl vs Unity application	112.5	0.002
	ArcGIS Online vs Unity application	295.0	0.692
C3 Terrain context	kepler.gl vs ArcGIS Online	32.5	below 0.001
	kepler.gl vs Unity application	26.5	below 0.001
	ArcGIS Online vs Unity application	235.0	0.348
C6 Attention held	kepler.gl vs ArcGIS Online	169.0	0.044
	kepler.gl vs Unity application	116.0	0.002
	ArcGIS Online vs Unity application	206.0	0.119

Table A4 Mean response per usability item, first tool encountered, on a scale from one to five

Item	kepler.gl	ArcGIS Online	Unity application
D1 Would like to use this frequently	3.33	3.92	3.30
D2 Unnecessarily complex	2.29	1.88	2.70
D3 Terrain data aided understanding	2.67	4.33	4.00
D4 Would need technical support	2.52	2.12	2.35
D5 Functions worked as expected	3.57	3.92	3.61
D6 Too much inconsistency	2.10	1.83	2.00
D7 Most people would learn it quickly	3.29	3.75	2.91
D8 Controls were easy to use	3.29	3.75	3.04
D9 Visualization performance was good	3.76	3.88	3.91
D10 Needed to learn a lot beforehand	2.24	1.96	2.26

Table A5 Measures pooled across all three viewing positions, 68 observations per tool

Measure	kepler.gl	ArcGIS Online	Unity application
Adapted usability score, mean	61.32	79.23	62.98
Adapted usability score, SD	18.13	14.96	16.56

Measure	kepler.gl	ArcGIS Online	Unity application
Rating mean, C1 to C6	3.40	4.13	4.06
Rating, SD	0.71	0.58	0.58
C1 Location could be identified	3.97	4.43	4.13
C2 Day-to-day change could be followed	3.85	4.43	4.10
C3 Terrain provided spatial context	2.78	4.21	4.41
C4 Detections distinguished from effects	3.53	3.91	3.99
C5 Meaning of marker differences was clear	3.10	3.91	3.49
C6 Attention was held throughout	3.19	3.91	4.22

Table A6 Counterbalancing scheme and realised group sizes

Version	Presentation order	Participants
ABC	kepler.gl, ArcGIS Online, Unity application	11
ACB	kepler.gl, Unity application, ArcGIS Online	10
BAC	ArcGIS Online, kepler.gl, Unity application	13

Version	Presentation order	Participants
BCA	ArcGIS Online, Unity application, kepler.gl	11
CAB	Unity application, kepler.gl, ArcGIS Online	12
CBA	Unity application, ArcGIS Online, kepler.gl	11

Table A7 Answer key for the scored comprehension items

Scored item	kepler.gl	ArcGIS Online	Unity application
Number of days displayed	Four	Six	Six
Detections on a stated date	50 to 99	100 to 149	Fewer than 50
Day of greatest activity	17 June	10 August	23 July
Terrain description	Mountain massif	Mountain massif	Mountain valley